A Stagewise Spectral-Structural Coordinated Reconstruction Network for Hyperspectral and Multispectral Image Fusion

Ruiping Liu, Jie Li, Pan Wang, and Chunyu Zhu

***Abstract*—Hyperspectral and multispectral image fusion (HMIF) aims to reconstruct high-resolution hyperspectral images (HR-HSIs) by integrating the rich spectral information of low-resolution hyperspectral images (LR-HSIs) with the fine spatial details of high-resolution multispectral images (HR-MSIs). In most existing methods, emphasis is still placed primarily on representation alignment, frequency enhancement, or progressive refinement, whereas the role of structural information throughout hyperspectral reconstruction has not been systematically distinguished with respect to its intervention timing, cross-modal transferability, and back-projection mechanism into the physical spectral-band domain. In particular, when cross-modal responses are inconsistent, spectral shifts and spurious textures may readily be induced by unconstrained or premature structural injection. To address this issue, a stagewise spectral-structural coordination network, termed SSCNet, is proposed. HMIF is reformulated as a constrained structural participation problem, and coordination between spectral preservation and spatial compensation is achieved through stagewise structural modulation. Specifically, SSCNet first constructs a hyperspectral-image-dominated (HSI-dominated) latent spectral basis; it then performs cross-modal screening of multispectral image (MSI) high-frequency structures under spectral guidance; finally, the screened structural information is progressively converted into reconstruction gains for HR-hyperspectral images (HSIs) through stagewise fusion back-projection. Experimental results on four datasets under 4×, 8×, and 16× degradation settings demonstrate that SSCNet achieves competitive performance across eight evaluation metrics, while exhibiting lower reconstruction error and reduced computational complexity. The code will be made available at https://github.com/LRuiRui517/SSCNetwork**

***Index Terms*—Cross-modal structural screening, hyperspectral and multispectral image fusion (HMIF), progressive fusion inference, spectral-structural collaborative reconstruction, stagewise fusion architecture.**

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Ruiping Liu is with the School of Information and Communication Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China, and also with the Ningbo Jiuzhang Zhitong Technology, Ningbo, Zhejiang 315207, China (e-mail: liuruiping@stu.xjtu.edu.cn).

Jie Li and Pan Wang are with the School of Information and Communication Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China (e-mail: jielixjtu@xjtu.edu.cn; panwang@stu.xjtu.edu.cn).

Chunyu Zhu is with Hangzhou Institute of Technology, Xidian University, Hangzhou, Zhejiang 311231, China (e-mail: zhuchunyu@xidian.edu.cn).

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I. Introduction

HYPERSPECTRAL images (HSIs), owing to their high-dimensional data representation and contiguous spectral sampling capabilities, enable precise characterization of material composition and subtle spectral variations and therefore are highly valuable for remote sensing tasks such as fine-grained classification, target recognition, change detection, and quantitative inversion [1], [2], [3]. However, because of limitations in sensor energy and imaging mechanisms, HSIs are often degraded by low spatial resolution, pronounced noise, and stripe artifacts. In contrast, multispectral images (MSIs) offer higher spatial resolution and better structural discriminability, but their utility is constrained by limited spectral dimensionality [4], [5]. Therefore, the fusion of low-resolution HSIs (LR-HSIs) with high-resolution MSIs (HR-MSIs) for the reconstruction of high-resolution HSIs (HR-HSIs) has become a significant research focus in remote sensing image fusion [6].

Hyperspectral and MSI fusion (HMIF) can be regarded as an ill-posed inverse problem under incomplete observations. The key challenge lies not only in simultaneously recovering spectral fidelity and spatial details but also in properly organizing the order and manner in which they participate and collaborate. Because different HSI bands exhibit substantial differences in signal-to-noise ratio (SNR), response range, and degree of degradation, direct cross-modal regression in the pixel domain, physical spectral-band domain, or shallow interaction stage, without the support of stable spectral representations, may cause structural compensation to prematurely interfere with spectral modeling [7], [8]. Moreover, owing to sparse spectral sampling, sensor-response discrepancies, and cross-modal inconsistencies, direct injection may also introduce pseudo-textures, local spectral distortion, and misleading details [9], [10]. Therefore, the difficulty of HMIF lies not only in jointly exploiting the two types of observations but also in avoiding excessive coupling between early spectral preservation and structural compensation and in hierarchically organizing the incorporation of structural information.

Existing HMIF methods can be broadly divided into two categories: traditional model-driven methods and deep learning-based methods. Traditional methods rely on explicit degradation models and prior constraints and therefore exhibit strong interpretability. However, they are prone to over-smoothing under complex textures and large-scale degradation, which makes it difficult to simultaneously preserve spatial details and spectral consistency [11], [12]. In recent years, deep learning methods have substantially improved HMIF performance. Related studies have gradually expanded from convolutional feature fusion to Transformer-based spectral–spatial dependency modeling, multiscale representation learning, and region-adaptive expert fusion. For example, recent methods such as spatial–spectral dilated transformer (SSDT) network [13] and region-aware MoE (RAMoE) [14] further enhance spectral–spatial representation and cross-modal fusion capability in complex scenes. However, in most deep methods, spectra and structures are still synchronously modeled in the pixel domain, physical spectral-band domain, or early feature-interaction stage, resulting in premature and strong coupling between the two [15]. Although latent-space modeling has been introduced in some methods, most studies still focus mainly on improving the utilization efficiency of cross-modal information, whereas insufficient attention has been paid to constraining the intervention of structural information in the spectral reconstruction process.

The above research progress indicates that the central issue in HMIF is no longer merely how low-resolution HSIs (LR-HSIs) and high-resolution MSIs (HR-MSIs) should be jointly exploited, but rather how the participation of structural information in hyperspectral reconstruction should be effectively regulated. Specifically, systematic modeling is still lacking in existing methods with respect to the following issues: how a stable spectral representation basis can be established before structural compensation, how the cross-modal transferability of high-frequency details from HR-MSIs can be assessed for HSI reconstruction, and how the screened structural information can be progressively back-projected into the physical spectral-band domain rather than being coupled into the final prediction in a one-shot manner. Based on these considerations, this study reformulates HMIF as a stagewise spectral-structural collaborative reconstruction problem with constrained structural participation and accordingly develops a stagewise spectral-structural coordination network (SSCNet). The main contributions of this study are summarized as follows.

1) A stagewise spectral-structural coordination network, termed SSCNet, is proposed. From the perspective of structural participation in the reconstruction process, this architecture formulates HR-HSI reconstruction as a progressive collaborative process that proceeds from spectral representation establishment and structural constraint introduction to reconstruction gain generation, thereby coordinating spectral preservation with spatial compensation.
2) A constrained structural-participation mechanism is constructed. Based on an HSI-dominated latent-space representation, frequency-response modeling and cross-modal consistency constraints are introduced to selectively exploit high-frequency structural information from HR-MSIs, thereby mitigating spectral distortions and spurious textures induced by cross-modal inconsistencies.
3) A progressive fusion-and-back-projection mechanism, termed the structure-constrained fusion inference (SCFI), is designed. SCFI formulates the screened structural information as a constrained structural prior, and the latent-space structural prior is progressively coordinated with the physical spectral-band reconstruction process through multistage residual inference, thereby improving the stability of spectral–spatial restoration under high-magnification degradation conditions.
4) SSCNet is systematically evaluated on four hyperspectral datasets under multiple degradation scales. Quantitative comparisons, visual analyses, bandwise evaluations, module ablation studies, and robustness experiments demonstrate that SSCNet achieves favorable reconstruction accuracy and a competitive performance-complexity tradeoff under a standard simulated HMIF protocol.

The remainder of this study is organized as follows. Section II introduces the related work. Section III presents the overall framework and key components of the proposed method. Section IV describes the experimental settings and analyzes the quantitative results and qualitative visual results of the method. Section V concludes this study and discusses future work.

II. Related Work

A. Fusion Methods Based on Physical Models and Priors

HMIF can usually be formulated as an ill-posed inverse problem constrained by spatial degradation and spectral response: HSI is mainly degraded by spatial blurring and downsampling, whereas MSI provides sparse observations of continuous spectra through spectral response functions [16]. Based on this formulation, explicit degradation models were commonly adopted in early studies and combined with prior constraints to construct optimization objectives, thereby improving the identifiability and stability of the solution [17]. Typical methods include subspace and low-rank modeling, in which the high-dimensional spectra of HSI are assumed to be approximately distributed in a low-dimensional subspace, and HR-HSI is represented as the product of a small number of basis vectors and spatial coefficients, thereby balancing computational efficiency and spectral consistency [18], [19]. In addition, sparse representation and dictionary learning methods are used to learn transferable sparse structures in the local patch domain, and high-resolution constraints from MSI are introduced to recover spatial details. In variational regularization and energy minimization methods, total variation, nonlocal similarity, or joint regularization terms are used to suppress noise and artifacts and enhance edge preservation [20]. The above methods exhibit good interpretability and physical consistency, but their performance is highly dependent on prior assumptions and weight selection. Under complex textures or large-scale degradation, these methods are prone to over-smoothing and detail loss; when cross-sensor response discrepancies are significant, the introduction of structural information may also induce local spectral distortion.

B. Deep Learning Methods

In recent years, substantial progress in HMIF has been achieved by deep learning methods, owing to their strong nonlinear representation and cross-modal modeling capabilities. According to their training paradigms and fusion mechanisms, existing deep learning methods can be broadly categorized into unsupervised observation-consistency methods, implicit- or latent-space representation methods, spatial–spectral interaction and unfolding-based methods, adaptive fusion methods, and spatial-frequency-domain guided methods. Although HSI-MSI fusion performance has been improved from different perspectives by these methods, their emphasis is still placed mainly on the utilization of cross-modal information, whereas systematic modeling remains lacking regarding how structural information is introduced, assessed for transferability, and back-projected during reconstruction [21].

In view of the difficulty associated with directly acquiring HR-HSIs in practical applications, unsupervised fusion methods have gradually attracted increasing attention. In such methods, observation-consistency constraints are typically constructed based on the imaging mechanism, such that the reconstructed results remain consistent with the input LR-HSI and HR-MSI after spatial degradation and spectral-response simulation, thereby enabling model learning without ground-truth HR-HSIs [22]. A generalizable unsupervised method was proposed by Yu et al. [23], in which spectral–spatial collaborative constraints are introduced, radiometric differences are mitigated through a group convolution enhancement (GCE), and feature extraction is enhanced using spatial, channel, and filter 3-D attention factor dynamic convolutional kernel (SCFConv). An unsupervised pretraining framework (UPFW) was proposed by Zheng et al. [24], in which low-resolution supervised pretraining is combined with full-resolution unsupervised adaptation to learn and transfer degradation parameters. Inverse mappings and degradation processes were learned by Jia et al. [25] through three-stage unsupervised pretraining, and cross-resolution feature extraction was optimized. Overall, unsupervised methods are more consistent with practical imaging conditions; however, their training signals are largely dependent on observation consistency. When degradation models, noise statistics, and cross-sensor radiometric differences cannot be accurately characterized, solution ambiguity may readily arise. This issue becomes particularly pronounced under high-magnification degradation and in complex texture regions, where structural restoration and spectral preservation remain strongly coupled.

In contrast, supervised learning remains an important paradigm for improving the performance ceiling of HSI-MSI fusion. In such methods, samples are typically constructed based on Wald's protocol [26] or synthetic degradation strategies, and end-to-end optimization is performed using ground-truth HR-HSIs as supervision, thereby directly constraining spectral fidelity and structural realism [27]. Within this paradigm, continuous spatial–spectral representation is enhanced in some methods through implicit representation or latent-space modeling. Implicit neural representation and Transformer self-attention were introduced by Zhu et al. [28], and a guided implicit sampling module was designed to alleviate the loss of spatial–spectral information caused by discrete explicit modeling and bilinear upsampling. A Mamba cooperative implicit neural representation fusion network (MCIFNet) was proposed by Zhu et al. [29], in which Mamba-based long-range dependency modeling is combined with continuous-domain representation, and point-to-point reconstruction is achieved through latent-space projection and a semantic extraction and fusion unit (SEFU), thereby improving spatial–spectral detail restoration. Overall, the limitations of direct pixel regression can be alleviated by latent-space or implicit representation. However, in existing methods, latent-space or implicit representation is primarily used for continuous representation, feature projection, or cross-modal alignment, whereas it has not yet been explicitly established as an HSI-dominated stable spectral basis prior to structural injection.

Another category of methods is focused on spatial–spectral correlation modeling and deep cross-modal interaction. A spectral dual adaptive graph embedding model (SDAGE) was proposed by Wang et al. [30], in which spatial nonlocal similarity (SNS) and spectral band correlation (SBC) are described by constructing spatial–spectral graphs, and these features are embedded to achieve fusion reconstruction under HR-HSI consistency constraints. A spatial–spectral unfolding network with mutual guidance (SMGU-Net) was proposed by Yan et al. [31], in which HR-MSI and LR-HSI are treated as complementary information sources, spatial–spectral correlations are explicitly modeled through a mutual-guidance reconstruction mechanism, and half-quadratic splitting is combined with gradient-descent-based optimization unfolding to achieve deep cross-modal interaction. In such methods, correlation modeling and iterative fusion between HSI and MSI are strengthened; however, emphasis is still placed primarily on enhancing the intensity of cross-modal interaction and optimization-based inference capability. The timing at which structural information intervenes in spectral reconstruction is usually not explicitly distinguished, nor is an independent assessment provided regarding whether different MSI high-frequency structures are suitable for transfer.

In addition, adaptive fusion and expert-learning methods are designed to dynamically adjust fusion strategies according to regional or modal discrepancies. An adaptive expert learning framework (AELF) was proposed by He et al. [32], in which modality dominance is modeled under regional variations in real-world scenes, bidirectional cross-attention between HSI and MSI is established through a modality-guided complementary module (MGCM) to extract multiscale complementary information, and features are decomposed into spatial, spectral, and edge subspaces through the attribute-aware mixture of fusion experts (AMoFEs) for soft-routing fusion. The limited representation capability of fixed fusion weights can be alleviated by such methods, and modality adaptability in complex scenes can be improved. However, structural information selection in these methods is mainly dependent on implicit network learning or expert routing allocation. High-frequency structural components from HR-MSIs have not yet been explicitly treated as candidate structural priors, nor has their cross-modal consistency with HSI spectral semantics been further assessed.

In recent years, frequency-domain information has been introduced into HMIF for enhanced edge and texture restoration. SFIGNet, a Kolmogorov–Arnold network (KAN)-based spatial-frequency dual-domain implicit guided sampling network, was proposed by Zhu et al. [33], in which implicit sampling and fusion decoding of HSI features are performed in the spatial and frequency domains through the spatial implicit guide sampling unit (SIGSU) and the frequency implicit guide sampling unit (FIGSU), respectively, thereby improving reconstruction performance while maintaining a lightweight parameterization. Spatial structural variations can be highlighted by frequency-domain modeling, which contributes positively to detail restoration in complex texture regions. However, in existing frequency-domain-guided methods, emphasis is mainly placed on exploiting frequency information to enhance sampling, decoding, or feature representation, whereas limited attention has been paid to whether MSI high-frequency responses necessarily correspond to reusable spatial structures in HR-HSIs.

In summary, important progress has been made by existing deep HMIF methods in feature representation, cross-modal interaction, and detail restoration, thereby continuously improving the performance of HSI-MSI fusion. However, a unified modeling framework remains lacking for characterizing when structural information should be introduced, how its transferability should be assessed, and how it should be back-projected during hyperspectral reconstruction.

C. Latent-Space Representation and Structural-Participation Constraints

Latent-space representation provides an effective strategy for alleviating the strong coupling induced by direct fusion in the pixel domain or the physical spectral-band domain. In related methods, high-dimensional HSIs are typically mapped into a more compact representation domain based on subspace assumptions, shared encoders, or implicit representation mechanisms, thereby reducing the difficulty of cross-modal fusion [34], [35]. However, existing latent spaces are primarily used for representation compression, modality alignment, or continuous spatial–spectral expression, whereas their functional role in the structural-participation process has not been sufficiently emphasized. Specifically, the latent space is usually not explicitly constructed in existing methods as an HSI-dominated stable spectral basis prior to structural injection, nor is the cross-modal transferability of high-frequency structural components from HR-MSIs explicitly assessed [36], [37]. Therefore, in scenes with complex textures, fragmented boundaries, or pronounced cross-sensor response discrepancies, structural information may still be injected into the reconstruction process implicitly or indiscriminately, thereby inducing local spectral shifts and spurious textures [38].

To address these issues, the latent space is further defined in this article as a unified carrier for spectral-basis construction, structural-transferability screening, and progressive back-projection. Unlike existing methods, in which the latent space is mainly used to improve fusion representation capability, greater emphasis is placed in SSCNet on constrained modeling of the entire process by which structural information participates in HR-HSI reconstruction. Therefore, the proposed method is not merely a latent-space alignment or frequency-domain enhancement approach, but rather a stagewise spectral-structural coordinated reconstruction framework for HMIF.

III. Methodology

A. Overall Framework

Let LR-HSI, HR-MSI, and the target HR-HSI be denoted by X_{LR}, X_{HR}, and X, respectively. Based on the above analysis, SSCNet is designed not merely to strengthen the interaction between HSI and MSI but to impose stagewise constraints on the entire process through which structural information is incorporated into HR-HSI reconstruction. The overall pipeline can be formulated as follows:

$$X_c = \Phi_{SCLM}(X_{LR}) \tag{1}$$

$$F_{MSI}^{fgsi} = \Phi_{FGSI}(F_{HSI}, X_{HR}) \tag{2}$$

$$\begin{aligned} X^0 &= X_c \\ \Delta X^{(t)} &= \Phi_{SCFI}\left(X^{(t)}, F_{MSI}^{fgsi}\right), \quad t = 0, 1, \ldots, T-1 \\ X^{(t+1)} &= X^{(t)} + \Delta X^{(t)} \\ X_{final} &= X^{(T)}. \end{aligned} \tag{3}$$

In conjunction with the above equations and Fig. 1(a), the spectral consistent latent modeling (SCLM) learns a spectrally consistent spectral–spatial latent representation F_{HSI} from the upsampled LR-HSI, which serves as the latent spectral basis for subsequent structural screening, and further decodes it to obtain the coarse HR-HSI prior X_c. The frequency-guided structure injection (FGSI) characterizes the consistency between the HSI latent feature F_{HSI} and the high-frequency structural responses of MSI in the shared latent space, and generates the structure-enhanced feature F_{MSI}^{fgsi} by selectively modulating MSI structural cues through a cross-modal gating mechanism. SCFI then organizes the spectral basis and structural prior into a constrained fusion state, and progressively maps this state back to the physical spectral domain in a multistage residual inference manner, ultimately yielding the reconstructed result X_{final}.

B. Spectral Basis Construction

As the first stage of SSCNet, SCLM aims to construct an HSI-dominated stable spectral basis prior to cross-modal structural interaction. In the HMIF task, if only the LR-HSI is interpolated and then directly used for pixel-level prediction, although the target resolution can be aligned at the spatial level, its representation still remains in the original physical spectral domain, which makes it difficult to stably capture high-dimensional spectral–spatial correlations and causes spectral shifts to accumulate during subsequent cross-modal interactions. To this end, this study designs SCLM, whose objective is not to directly regress the final HR-HSI from the upsampled input, but rather to first construct a spectral–spatial latent representation with spectral consistency in a shared latent space, and to use this representation as a unified spectral basis for subsequent structure selection and progressive reconstruction. The input LR-HSI is denoted by $X_{LR} \in \mathbb{R}^{B\times C_h\times h\times w}$, where B denotes the batch size, C_h denotes the number of HSI bands, and h and w denote the spatial height and width of the low-resolution input, respectively. First, the input is mapped to the target spatial scale via the upsampling operator to obtain $X_{LR}^{\uparrow} \in \mathbb{R}^{B\times C_h\times H\times W}$, where H and W denote the target

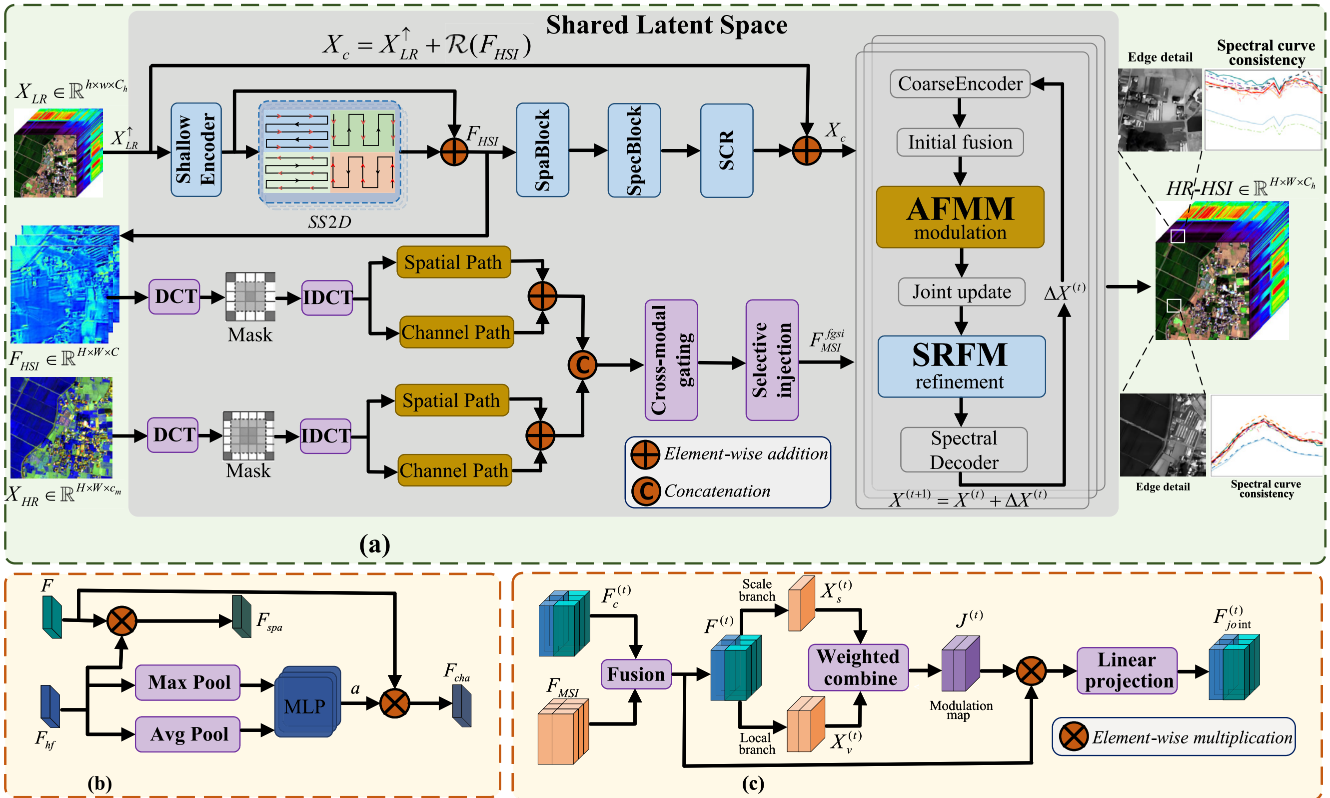

Fig. 1. Overall architecture and key modules of SSCNet. (a) Stagewise spectral-structural coordination framework, including SCLM, FGSI, and SCFI. (b) Spatial path and channel path in FGSI. (c) AFMM in SCFI.

high-resolution spatial dimensions, respectively. Subsequently, SCLM adopts a residual formulation to construct a coarse HR-HSI prior

$$X_c = X_{LR}^{\uparrow} + \mathcal{R}(F_{HSI}) \tag{4}$$

where $X_c \in \mathbb{R}^{B \times C_h \times H \times W}$ denotes the coarse HR-HSI prior output by SCLM; F_{HSI} denotes the spectral–spatial joint latent representation learned in the shared latent space; and $\mathcal{R}(\cdot)$ denotes the residual generation operator, which is used to map the latent-space features to a compensatory residual in the physical spectral domain.

After obtaining $X_{LR}^{\uparrow}$, SCLM first compresses the original spectral bands into a set of unified representation channels through a shallow encoder $\varepsilon_0(\cdot)$, yielding the initial spectral–spatial features

$$F_0 = \varepsilon_0\left(X_{LR}^{\uparrow}\right) \tag{5}$$

where $F_0 \in \mathbb{R}^{B \times C \times H \times W}$, and C denotes the number of feature channels in the unified latent space. To more comprehensively characterize the cross-spatial-scale and cross-band correlations in HSI, SCLM stacks K deep blocks based on 2-D selective scan (SS2D). Let the input feature of the kth deep block be denoted as F_k, and its update rule is defined as

$$F_{k+1} = F_k + \mathcal{S}(F_k), \quad k = 0, \cdots K-1 \tag{6}$$

where $\mathcal{S}(\cdot)$ denotes the SS2D-based spectral–spatial modeling operator. This operator first performs linear projection and normalization along the channel dimension, then aggregates local context through depthwise separable convolution, and subsequently conducts state-space scanning along four directions, namely, the horizontal and vertical directions and their respective reverse directions, followed by the fusion of the responses from different directions. After stacking K deep blocks, the unified shared latent spectral–spatial feature can be obtained as

$$F_{HSI} = F_K \in \mathbb{R}^{B \times C \times H \times W}. \tag{7}$$

F_{HSI} denotes the HSI latent feature learned by the LR-HSI branch. This feature does not directly correspond to the final output, but instead serves two roles: first, it provides a spectrally consistent prior for coarse HR-HSI reconstruction; and second, it serves as a unified latent representation basis shared by the subsequent cross-modal structure selection module. To further improve the response to detailed regions, SCLM introduces a lightweight spatial refinement operator, SpaBlock, in the shared latent space to perform shallow residual enhancement on F_{HSI}, yielding

$$F_{sp} = F_{HSI} + \varepsilon_{spa}(F_{HSI}) \tag{8}$$

where $\varepsilon_{spa}(\cdot)$ denotes the shallow spatial residual refinement operator, and $F_{sp} \in \mathbb{R}^{B \times C \times H \times W}$ denotes the latent feature after local spatial enhancement. Subsequently, SCLM maps the latent feature back to the physical spectral domain through the spectral decoder SpecBlock to generate the initial residual

$$R_{raw} = \mathcal{D}_{spec}\left(F_{sp}\right) \tag{9}$$

where $\mathcal{D}_{spec}(\cdot)$ denotes the spectral decoding operator, and $R_{raw} \in \mathbb{R}^{B\times C_h\times H\times W}$ denotes the initial bandwise compensatory residual decoded from the shared latent representation. Considering that different bands in real HSI exhibit substantial differences in SNR, radiometric response, and information content, this study further introduces a spectral channel recalibration (SCR) module to perform channel-wise modulation on R_{raw}, yielding

$$R = \mathcal{C}_{scr}\left(R_{raw}\right) \tag{10}$$

where $\mathcal{C}_{scr}(\cdot)$ denotes the SCR operator, and $R \in \mathbb{R}^{B\times C_h\times H\times W}$ denotes the residual result after band reliability correction. Substituting (10) into (4), the coarse HR-HSI prior X_c output by SCLM can be obtained. Therefore, SCLM does not directly undertake the final reconstruction task, but instead constructs an HR-HSI prior with both spectral stability and preliminary spatial representational capacity through spectral–spatial consistent modeling, local spatial refinement, spectral decoding, and band recalibration in the shared latent space, thereby providing a unified and reliable representation basis for subsequent transferable structure selection and progressive fusion inference.

C. Cross-Modal Structural Screening

In the second stage of SSCNet, FGSI further addresses the issues of whether structural information should be injected and how it should be injected. Although F_{HSI} output by SCLM has good spectral consistency, it is derived from the upsampled LR-HSI, and the high-resolution structural details that it can provide are limited. In contrast, the original MSI contains richer spatial edges and texture information, but lacks the full representational capacity of HSI. Therefore, the purpose of FGSI is not to simply inject the high-frequency information from MSI directly into the HSI branch, but to suppress indiscriminate structural injection by screening MSI high-frequency responses under cross-modal guidance and incorporating them in a controlled manner. Based on this idea, FGSI treats high-frequency structures as candidate detail information rather than implicitly assuming them to be valid priors, and generates structure-enhanced features coordinated with the HSI spectral basis through frequency-aware perception and a gated screening mechanism, denoted as F_{MSI}^{fgsi}, for subsequent use by SCFI in the projection back to the physical spectral domain. The input HR-MSI is denoted by $X_{HR} \in \mathbb{R}^{B\times C_m\times H\times W}$, where C_m denotes the number of MSI bands. To maintain consistency with the HSI latent feature $F_{HSI} \in \mathbb{R}^{B\times C\times H\times W}$ output by SCLM, FGSI first projects the original MSI into the unified latent space through a mapping operator $\varepsilon_m(\cdot)$, yielding

$$F_{MSI} = \varepsilon_m\left(X_{HR}\right) \tag{11}$$

where $F_{MSI} \in \mathbb{R}^{B\times C\times H\times W}$ denotes the channel-aligned MSI latent feature, and $\varepsilon_m(\cdot)$ denotes a modality alignment operator composed of linear mapping and local spatial encoding. As shown in Fig. 1(b), the single-modal high-frequency perception module within FGSI consists of a spatial path and a channel path, both of which follow a high-pass filtering strategy based on the 2-D discrete cosine transform (DCT). Given any input feature $F \in \mathbb{R}^{B\times C\times H\times W}$, a 2-D DCT is first performed on each channel along the spatial dimension to obtain the frequency-domain coefficients

$$F_{dct} = \mathcal{DCT}\left(F\right) \tag{12}$$

$\mathcal{DCT}(\cdot)$ denotes the 2-D DCT operator. Subsequently, a high-frequency mask is constructed according to the hyperparameter pair (r_h, r_w)

$$M\left(i, j\right) = \begin{cases} 0, & i < r_h H,\ j < r_w W \\ 1, & otherwise \end{cases} \tag{13}$$

where $M \in \{0, 1\}^{H\times W}$, and r_h and r_w, respectively, control the proportion of the low-frequency removal region along the height and width directions. The zeroed region in the upper left corner corresponds to the low-frequency components. After removing the low-frequency components with this mask and transforming the result back to the spatial domain through the inverse DCT (IDCT), the high-frequency response can be obtained as

$$F_{hf} = \mathcal{IDCT}\left(M \odot F_{dct}\right) \tag{14}$$

where $\mathcal{IDCT}(\cdot)$ denotes the 2-D IDCT operator, $\odot$ denotes element-wise multiplication, and the mask M is broadcast along the batch and channel dimensions. Since DCT has the property of energy compaction, the above masking operation is equivalent to suppressing the low-frequency components that dominate the signal energy, thereby highlighting the high-frequency information related to edges and textures.

In the spatial path, the high-frequency response is regarded as a spatial modulation factor for the original feature and acts on the input feature in an element-wise manner, thereby achieving explicit enhancement of edges and textures

$$F_{spa} = F \odot F_{hf} \tag{15}$$

where $F_{spa} \in \mathbb{R}^{B\times C\times H\times W}$ denotes the spatial-path feature after high-frequency modulation. While preserving the original structural distribution, this process enhances the response intensity in high-frequency-rich regions, enabling the spatial path to pay more attention to local edges and texture variations. The channel path adaptively estimates channel weights on the basis of frequency statistics. Specifically, adaptive max pooling and average pooling are applied to F_{hf} in (14), respectively, to obtain feature blocks with a size of (h_p, w_p)

$$\begin{aligned} F^{max} &= \mathcal{AP}_{max}\left(F_{hf}\right) \\ F^{avg} &= \mathcal{AP}_{avg}\left(F_{hf}\right) \end{aligned} \tag{16}$$

where $\mathcal{AP}_{max}(\cdot)$ and $\mathcal{AP}_{avg}(\cdot)$ denote the adaptive max pooling and adaptive average pooling operators, respectively, and (h_p, w_p) is specified by the hyperparameter patch size, which is used to control the spatial range over which the frequency response is aggregated. Then, nonlinear activation and summation are performed within the local block for each channel, yielding two complementary channel statistics vectors

$$\begin{aligned} g^{max} &= sum\left(\sigma\left(F^{max}\right)\right) \\ g^{avg} &= sum\left(\sigma\left(F^{avg}\right)\right) \end{aligned} \tag{17}$$

where $g^{max}, g^{avg} \in \mathbb{R}^{B\times C\times 1\times 1}$, $\sigma(\cdot)$ denotes the nonlinear activation function, and $sum(\cdot)$ denotes summation within the pooling block region. Max pooling emphasizes the strongest local high-frequency response, whereas average pooling reflects the overall high-frequency energy. Their combination can more comprehensively capture the importance of each channel in high-frequency structures. Next, a lightweight channel MLP is constructed to fuse the two statistical descriptors and generate the channel attention weights

$$s = \delta\left(W_1\left(g^{max} + g^{avg}\right)\right) \tag{18}$$

$$a = \sigma\left(W_2\left(s\right)\right) \tag{19}$$

where $W_1(\cdot)$ and $W_2(\cdot)$ denote the two linear mapping layers, $\delta(\cdot)$ denotes the activation function, and $a \in \mathbb{R}^{B\times C\times 1\times 1}$ denotes the channel attention weights. This design controls the number of parameters and computational cost while maintaining relatively high channel-modeling efficiency under high-resolution inputs. Accordingly, the output of the channel path is

$$F_{cha} = F \odot a \tag{20}$$

where $F_{cha} \in \mathbb{R}^{B\times C\times H\times W}$ denotes the feature after channel reweighting, which is used to highlight the channels with stronger responses to high-frequency structures while suppressing noisy and redundant channels.

In single-modal FGSI, the two features obtained from the spatial path and the channel path are added and then unified through linear mapping, yielding the single-modal high-frequency encoding result

$$F^{out} = \psi\left(F_{spa} + F_{cha}\right) \tag{21}$$

where $\psi(\cdot)$ denotes a linear mapping operator, which is used to unify the output distributions of the spatial path and the channel path. The same form is adopted for both the HSI branch and the MSI branch, so that comparable high-frequency structural features can be extracted in the shared latent space, thereby providing a consistent frequency characterization basis for cross-modal interaction. Based on this, the single-modal high-frequency encoding operator $\mathcal{H}(\cdot)$ is defined as the overall combination of the above frequency perception, spatial path, and channel path. Then, the high-frequency structural features of the HSI and MSI branches are, respectively, obtained as

$$\begin{aligned} F^{out}_{HSI} &= \mathcal{H}\left(F_{HSI}\right) \\ F^{out}_{MSI} &= \mathcal{H}\left(F_{MSI}\right). \end{aligned} \tag{22}$$

Since the two modalities complete high-frequency encoding under the same structural form, F^{out}_{HSI} and F^{out}_{MSI} are consistent in frequency bandwidth and statistical characterization, which enables the high-frequency structural representations of the two modalities to be concatenated along the channel dimension in the cross-modal modeling stage

$$F_{cat} = Cat\left(F^{out}_{HSI}, F^{out}_{MSI}\right) \tag{23}$$

where $Cat(\cdot)$ denotes the channel concatenation operation. Subsequently, a position-dependent gating map is constructed through the gate generation operator $\mathcal{G}(\cdot)$

$$G = \sigma\left(\mathcal{G}\left(F_{cat}\right)\right) \tag{24}$$

where $G \in \mathbb{R}^{B\times C\times H\times W}$ denotes the cross-modal gating map, and $\mathcal{G}(\cdot)$ is composed of a linear mapping and a nonlinear activation, which is used to generate position-wise and channel-wise selective weights according to the consistency of high-frequency responses from the two modalities. It should be noted that the gating map G is not merely used to measure the strength of high-frequency responses at different positions, but more importantly, to generate position-wise and channel-wise modulation weights from cross-modal structural consistency, thereby regulating the degree to which MSI high-frequency cues participate in subsequent reconstruction. Finally, the structure-enhanced MSI feature output by cross-modal FGSI is defined as

$$F^{fgsi}_{MSI} = F_{MSI} + G \odot F^{out}_{MSI}. \tag{25}$$

Through this additive residual form, the network can automatically balance the relative contributions of the original MSI feature and the high-frequency compensation term during training, thereby preserving stable structural information in the high-resolution image while enhancing detail representation in a manner coordinated with spectral preservation under the guidance of HSI. Accordingly, FGSI achieves selective injection of cross-modal high-frequency structures in the shared latent space, enabling the enhanced MSI structural feature F^{fgsi}_{MSI} to be coordinated more naturally with the HSI spectral basis and providing a more reliable structural prior for subsequent reconstruction.

D. Progressive Fusion Inference

After spectral basis construction and transferable structure screening are completed, the third stage, SCFI, organizes the two types of information into a constrained fusion state and progressively transforms them into the final reconstruction result. In SSCNet, SCLM provides a coarse high-resolution estimate X_c with spectral consistency and its corresponding shared latent representation, whereas FGSI generates the structure-enhanced feature F^{fgsi}_{MSI} that is coordinated with this spectral basis. However, these two types of information still exhibit inherent differences in representational emphasis: the former places greater emphasis on spectral stability, whereas the latter focuses more on spatial detail restoration. If the two are directly fused once in the pixel domain, this may still lead to excessive structural compensation or the accumulation of local spectral deviations. Based on this, SCFI is proposed. Its core function is not residual refinement in the general sense, but rather to organize the spectral basis and structural prior in the shared latent space into a constrained fusion state, and to progressively project this state back to the physical spectral domain through multistage inference. First, in order to obtain a fused basis with both spectral consistency and preliminary spatial representation capability at each inference stage, a CoarseEncoder is designed to perform spectral encoding on the current estimate $X^{(t)}$. Specifically, the original HSI bands are compressed into a unified fusion dimension through the initial encoding operator $\phi_s(\cdot)$, yielding

$$U^{(t)}_0 = \phi_s\left(X^{(t)}\right) \tag{26}$$

where the operator $\phi_s(\cdot)$ consists of spectral linear mapping and local spatial enhancement. On this basis, K_c local mixing blocks are further stacked to progressively enhance the spectral–spatial features. The update rule of the kth block is defined as

$$U_{k+1}^{(t)} = U_k^{(t)} + \gamma_k \mathcal{P}\left(U_k^{(t)}\right), \quad k = 0, 1, \ldots, K_c - 1 \tag{27}$$

where $\mathcal{P}(\cdot)$ denotes the local structure enhancement operator, which is used to mix spatial information and channel information within a local neighborhood, and γ_k is the learnable scaling coefficient corresponding to the kth residual block, which is used to control the residual magnitude and avoid feature oscillation during deep iterative processing. To further enhance the spectral correlation among channels, a lightweight spectral MLP residual is introduced at the output end, yielding the coarse encoded feature at the current stage

$$F_c^{(t)} = U_{K_c}^{(t)} + \gamma_{spec}\mathcal{M}_{spec}\left(U_{K_c}^{(t)}\right) \tag{28}$$

where $\mathcal{M}_{spec}(\cdot)$ denotes the lightweight spectral MLP operator, and γ_{spec} is the corresponding residual scaling coefficient. In this way, the CoarseEncoder outputs at each inference stage a feature representation with relatively strong spectral consistency and preliminary spatial representational capacity, thereby providing stable input for subsequent structural modulation. As shown in Fig. 1(c), after obtaining $F_c^{(t)}$, the adaptive feature modulation module (AFMM) is required to further introduce the structural prior F_{MSI}^{fgsi} into the fusion process of the current stage. To this end, the two are first fused after channel alignment, yielding the intermediate feature

$$f^{(t)} = \phi_f\left(F_c^{(t)}, F_{MSI}^{fgsi}\right) \tag{29}$$

where $\phi_f(\cdot)$ denotes the channel alignment and initial feature fusion operator, and $f^{(t)}$ denotes the basic fused representation prior to structural modulation, which is subsequently used for spatial structure characterization and local constraint modeling. To characterize the structural distribution properties of this feature, a scale-structure term and a structure-strength term are constructed, respectively. The scale-structure term is defined as

$$x_s^{(t)} = \mathcal{P}_s\left(f^{(t)}\right) \tag{30}$$

where $\mathcal{P}_s(\cdot)$ denotes the spatial compression and structure extraction operator, which is used to capture the scale distribution information of the feature over a larger receptive field. The structure-strength term is formulated based on local variance statistics

$$x_v^{(t)} = \sqrt{Var_{h_a,w_a}\left(f^{(t)}\right) + \varepsilon} \tag{31}$$

where $Var_{h_a,w_a}(\cdot)$ denotes the variance computed within a local neighborhood with spatial window size $h_a \times w_a$, which is used to characterize the magnitude of local texture fluctuations; and ε is a numerical stabilization constant introduced to prevent training instability resulting from excessively small or zero variance values. Based on the two complementary structural descriptions, a structural modulation map is further generated through linear weighting, nonlinear activation, and scale restoration

$$J^{(t)} = Up\left[\sigma\left(\alpha x_s^{(t)} + \beta x_v^{(t)}\right)\right] \tag{32}$$

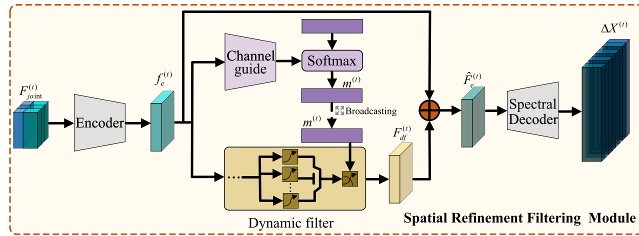

Fig. 2. Architecture of the SRFM.

where $J^{(t)} \in \mathbb{R}^{B\times C\times H\times W}$, α and β are learnable parameters, and $Up[\cdot]$ denotes the upsampling or scale restoration operation. Here, the role of $J^{(t)}$ is not to directly generate fine details but to explicitly regulate where and to what extent the structural prior should be introduced: in regions with complex textures and significant local fluctuations, the modulation weight is higher; in smooth regions, a more conservative compensation strategy is adopted to reduce the risk of spectral distortion caused by incorrect structural guidance. After obtaining the structural modulation map $J^{(t)}$, AFMM employs it to modulate the basic fused feature $f^{(t)}$ in an element-wise manner, and constrained feature updating is performed through linear mapping, yielding

$$F_{joint}^{(t)} = f^{(t)} + \psi_j\left(f^{(t)} \odot J^{(t)}\right) \tag{33}$$

where $F_{joint}^{(t)} \in \mathbb{R}^{B\times C\times H\times W}$, and $\psi_j(\cdot)$ denotes linear mapping. This design enables the fusion result to rely more heavily on the structural prior in texture-rich regions while preserving spectral dominance in smooth regions, thereby achieving modality-adaptive feature fusion under structural constraints. As shown in Fig. 2, to further transform structural constraints in the latent space into effective refined residuals in the spatial domain, a spatial refinement filtering module (SRFM) is introduced for dynamic restoration modeling of $F_{joint}^{(t)}$. First, the basic spatial representation is extracted through the encoder $\varepsilon_r(\cdot)$

$$F_e^{(t)} = \varepsilon_r\left(F_{joint}^{(t)}\right) \tag{34}$$

where $F_e^{(t)} \in \mathbb{R}^{B\times C\times H\times W}$ denotes the basic spatial representation at the current stage. Subsequently, a global channel guidance vector is derived from $F_e^{(t)}$

$$m^{(t)} = Softmax\left(d\left(F_e^{(t)}\right)\right) \tag{35}$$

where $m^{(t)} \in \mathbb{R}^{B\times C\times 1\times 1}$, and $d(\cdot)$ denotes a global channel description operator, which is used to extract the response preference of different channels at the current stage from an overall perspective. $Softmax(\cdot)$ is used to normalize these responses into dynamic filtering control weights. After $m^{(t)}$ is broadcast along the spatial dimensions, it is denoted as $\tilde{m}^{(t)} \in \mathbb{R}^{B\times C\times H\times W}$, which serves as the control signal for the dynamic filtering operator and is involved in subsequent spatial refinement. Based on this, the refined response is generated by the dynamic filtering operator

$$F_{df}^{(t)} = \mathcal{F}_{df}\left(F_e^{(t)}, \tilde{m}^{(t)}\right) \tag{36}$$

TABLE I
EXPERIMENTAL DATASETS

Datasets	DFC2018	Chikusei	PaviaC	PaviaU
HSI bands	48	128	102	103
MSI bands	4	4	4	4
Spatial resolution (m)	1	2.5	1.3	1.3
Spectral range (nm)	380-1050	343-1018	430-860	430-860
Size	1202×4172	2517×2335	1096×1096	610×340

where $\mathcal{F}_{df}(\cdot)$ denotes a dynamic filtering operator composed of multiscale convolution branches and an adaptively weighted combination thereof, whose branch weights are learned under the control of the guidance information $\tilde{m}^{(t)}$. By adding the dynamic filtering result to the basic spatial feature in a residual manner, the enhanced spatial representation is obtained

$$\hat{F}_e^{(t)} = F_e^{(t)} + F_{df}^{(t)}. \tag{37}$$

Finally, the enhanced representation is mapped back to the physical spectral domain through the spectral decoder $\mathcal{D}_s(\cdot)$ to produce the reconstruction residual at the current stage

$$\Delta X^{(t)} = \mathcal{D}_s\left(\hat{F}_e^{(t)}\right) \tag{38}$$

which is then substituted into (3) to complete the update of the HR-HSI estimate at the tth stage.

Through the above mechanism, SCFI no longer treats reconstruction as a one-shot output, but models it as a progressive projection process from the shared latent representation to the physical spectral domain. CoarseEncoder provides a spectrally consistent fusion basis at each stage, AFMM adaptively modulates the contribution strength of the structural prior according to the local structural distribution, and SRFM further transforms the constrained latent fusion state into refined residuals in the physical spectral domain. In this way, SCFI can more stably balance structural enhancement and spectral preservation than one-shot residual prediction and, together with SCLM and FGSI, forms the stagewise spectral-structural collaborative reconstruction framework of SSCNet.

E. Loss Function

To ensure training stability and avoid additional uncertainty caused by the coupling of weights across complex loss terms, the L_1 loss is adopted in this study as the basic training objective. Compared with the L_2 loss, the L_1 loss exhibits a more consistent gradient response when penalizing both large and small errors, which helps alleviate over-smoothing and preserve edge and structural details. It is formulated as

$$Loss = \frac{1}{n}\sum_{i=1}^{n}\left|out_i - ref_i\right| \tag{39}$$

where out_i denotes the predicted value of the ith pixel in the output image, ref_i denotes the pixel value of the reference image at the corresponding position, and n denotes the total number of pixels in the image.

IV. EXPERIMENTS

A. Dataset Description

To systematically investigate the applicability and performance of SSCNet in multiple scenarios, four representative hyperspectral datasets are selected for evaluation: DFC2018, Chikusei, PaviaC, and PaviaU. These datasets exhibit significant differences in spatial and spectral scales, thereby enabling the fusion performance and robustness of the proposed model to be evaluated under complex data distributions from multiple perspectives. Specifically, the DFC2018 dataset was collected using the ITRES CASI-1500 airborne sensor and was released as an open-access dataset for the 2018 IEEE GRSS Data Fusion Contest, with annotations for 20 land-cover categories. The Chikusei dataset was acquired by the Headwall Hyperspec-VNIR-C system in the summer of 2014, covering the Chikusei area of Ibaraki Prefecture, Japan, and contains annotations for 19 typical land-cover classes. Both PaviaC and PaviaU were acquired by the ROSIS system in northern Italy, and after the removal of invalid bands, each dataset retains nine land-cover categories. Comparative experiments on these datasets, which differ significantly in imaging mode, resolution, and scene characteristics, facilitate a comprehensive evaluation of the adaptability and performance stability of SSCNet under diverse data distributions. The specific parameters of the datasets are summarized in Table I.

B. Experimental Settings and Implementation Details

All experiments in this study were conducted in a Python 3.10 environment, with PyTorch 2.1 adopted as the deep learning framework. In terms of hardware configuration, an NVIDIA 4060 Ti GPU with 16 GB of memory was used to ensure efficient model training and inference. To construct the multisource data required for the experiments, Wald's protocol was followed for simulation: the original HSI was used as the high-quality reference, and the corresponding HSI and MSI data for training and testing were generated through downsampling in both the spatial and spectral dimensions. Specifically, let the spectral response matrix be $R \in \mathbb{R}^{C_h \times C_m}$. To ensure that different MSI bands have a consistent response energy scale, this article normalizes R column-wise, so that $\sum_{b=1}^{C_h} R_{b,m} = 1$. Therefore, X_{HR} generated from the reference X can be expressed as

$$X_{HR}^{(m)}(i,j) = \sum_{b=1}^{C_h} X^{(b)}(i,j)\, R_{b,m}, \quad m = 1, \ldots, C_m \tag{40}$$

where $R_{b,m}$ denotes the response weight from the bth HSI band to the mth MSI band. The spectral response matrix

TABLE II
RESULTS ON THE DFC2018 DATASET

Ratio	Metrics	CNMF	GSA	HySure	ITF-GAN	SDAGE	MCIFNet	SMGU-Net	AELF	SFIGNet	SSCNet
×4	PSNR↑	42.6061	46.4165	44.0793	46.8118	41.6489	47.7069	47.9272	43.0008	47.9505	**50.6844**
	RMSE↓	1.3136	0.8471	1.1087	1.0086	1.8275	0.9098	0.8871	1.5641	0.8847	**0.6458**
	ERGAS↓	1.0305	0.6229	1.0879	0.8006	1.3116	0.7539	0.7137	0.8908	0.8226	**0.5005**
	SAM↓	1.1279	1.1194	1.3354	1.2678	2.2446	1.1724	1.1574	1.7522	1.1192	**0.8093**
	CC→1	0.9996	0.9998	0.9997	0.9998	0.9992	0.9998	0.9998	0.9995	0.9998	**0.9999**
	MoAE↓	0.9611	0.5571	0.7865	0.6923	1.3461	0.5942	0.5854	1.0742	0.5317	**0.4057**
	UIQI→1	0.9945	0.9983	0.9938	0.9979	0.9957	0.9981	0.9982	0.9980	0.9974	**0.9992**
	SSIM↑	0.9908	0.9927	0.9909	0.9960	0.9895	0.9968	0.9969	0.9940	0.9966	**0.9979**
×8	PSNR↑	37.4372	39.4609	39.1002	45.9518	42.2402	46.9261	46.9841	46.9753	46.9027	**48.3471**
	RMSE↓	2.3819	1.8869	1.9894	1.1136	1.7073	0.9954	0.9888	0.9898	0.9981	**0.8452**
	ERGAS↓	0.8625	0.5826	0.8057	0.4770	0.6187	0.4162	0.4873	0.3598	0.4602	**0.3277**
	SAM↓	1.8610	2.2377	1.9934	1.4232	1.9439	1.2858	1.2745	1.2134	1.2595	**1.0485**
	CC→1	0.9988	0.9987	0.9992	0.9997	0.9994	0.9998	0.9998	0.9998	0.9998	**0.9998**
	MoAE↓	1.8162	1.3757	1.4649	0.7724	1.2409	0.6510	0.6511	0.6608	0.5923	**0.5401**
	UIQI→1	0.9885	0.9961	0.9906	0.9969	0.9956	0.9976	0.9971	0.9984	0.9967	**0.9987**
	SSIM↑	0.9752	0.9841	0.9849	0.9953	0.9912	0.9964	0.9964	0.9965	0.9960	**0.9971**
×16	PSNR↑	24.2164	26.4200	27.9458	45.7742	41.8280	46.7622	46.7717	45.5557	46.1359	**47.3381**
	RMSE↓	10.9130	8.4680	7.1038	1.1366	1.7902	1.0144	1.0133	1.1655	1.0902	**0.9493**
	ERGAS↓	1.4757	1.1508	0.9815	0.2734	0.2883	0.2313	0.2409	0.2273	0.2568	**0.1902**
	SAM↓	4.4082	9.1813	4.1476	1.4726	1.9803	1.2897	1.2982	1.4430	1.3465	**1.1791**
	CC→1	0.9605	0.9813	0.9893	0.9997	0.9993	0.9998	0.9998	0.9997	0.9997	**0.9998**
	MoAE↓	7.2913	5.9587	4.6156	0.7823	1.2917	0.6615	0.6669	0.7988	0.6504	**0.6241**
	UIQI→1	0.9460	0.9682	0.9738	0.9955	0.9960	0.9969	0.9968	0.9973	0.9957	**0.9980**
	SSIM↑	0.7649	0.8985	0.9118	0.9953	0.9908	0.9963	0.9962	0.9958	0.9956	**0.9967**

R was read from the corresponding ∗_SRF.xls file of each dataset and normalized column-wise. Spatial degradation is then simulated using Gaussian blurring. The Gaussian kernel size is set to $\lambda \times \lambda$, where λ denotes the degradation scale factor; the standard deviation of the Gaussian kernel is fixed at $\eta = 2$, and the downsampling stride is kept identical to the scale factor λ. For dataset partitioning, a fixed 128×128 region was cropped from each image as the test set and excluded from the training image to ensure strict separation between the training and testing data. During training, image patches were extracted from the remaining regions according to a predefined coordinate sequence and used for network optimization, whereas the fixed held-out region was used only for convergence monitoring and checkpoint saving. All major training settings were fixed before the experiments and kept consistent for all deep learning methods. To ensure a fair comparison among different deep models, the experimental strategy was uniformly configured as follows: the total number of training iterations was 10 000; the learning rate was set to 1×10^{-4}, and AdamW was adopted as the optimizer to ensure stable convergence behavior.

C. Evaluation Metrics

To more comprehensively and objectively evaluate the performance of each fusion method in the reconstruction task, eight commonly used quantitative metrics were selected for the simulated-data experiments, including peak SNR (PSNR) [39], root-mean-square error (RMSE) [40], Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [41], spectral angle mapper (SAM) [42], correlation coefficient (CC) [43], mean absolute error (MoAE) [44], universal image quality index (UIQI) [45], and structural similarity (SSIM) [46]. Among them, PSNR, RMSE, and MoAE mainly reflect reconstruction error; SAM, ERGAS, and CC focus on spectral consistency; and UIQI and SSIM are used to measure spatial structure and overall quality, thereby providing a systematic and comprehensive evaluation of the fusion results.

D. Experimental Results and Analysis

To comprehensively validate the effectiveness and advantages of SSCNet, multiple representative baseline methods are selected, including both traditional approaches and deep learning-based models. The traditional approaches include the image-enhancement-based GSA [11] and the decomposition-based CNMF [12] and HySure [17]. The deep learning-based methods include several representative models proposed in recent years, namely, ITF-GAN [28], SDAGE [30], MCIFNet [29], SMGU-Net [31], AELF [32], and SFIGNet [33]. Experiments are conducted under three degradation scale factors, i.e., 4×, 8×, and 16×, and the results are evaluated in terms of spectral fidelity, spatial detail restoration, and overall reconstruction quality. Error maps are further used to analyze local reconstruction deviations. In the result tables, the best, second-best, and third-best results are highlighted in red, blue, and green, respectively, to facilitate an intuitive comparison of performance differences among the methods.

1) Analysis of the DFC2018 dataset: DFC2018 is characterized by complex urban structures, such as dense buildings, road intersections, and roof edges, which impose stringent requirements for spatial detail restoration and spectral preservation. As shown in Table II, as the degradation scale increases, traditional methods generally exhibit edge blurring and local structural loss, whereas leading performance is maintained by SSCNet across all three scale factors. In the visualization results shown in Fig. 3, road contours and building boundaries are restored more clearly and completely,

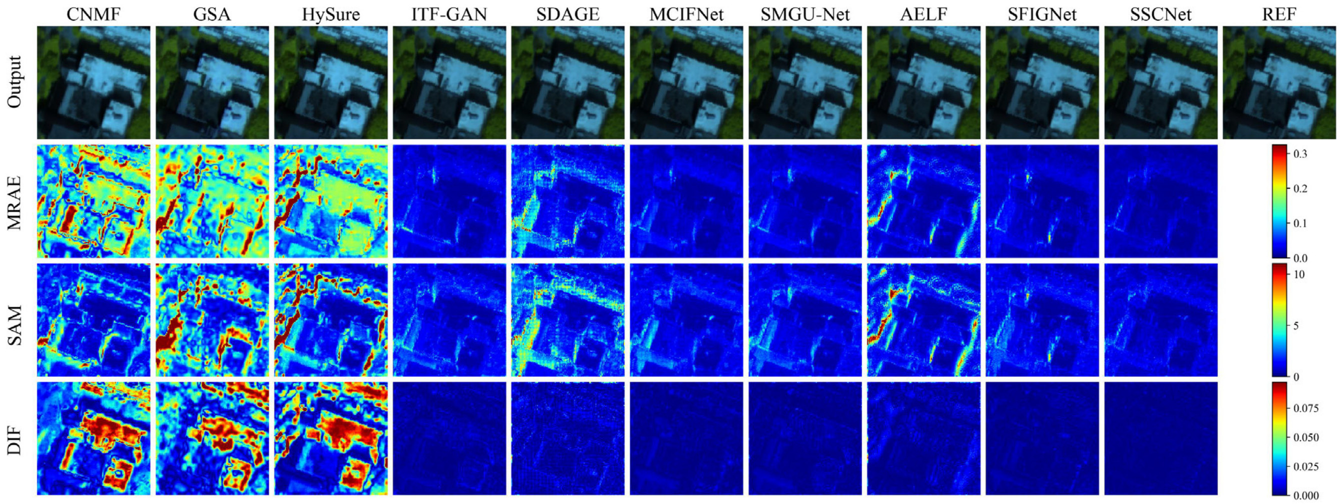

Fig. 3. Experimental results on the DFC2018 dataset (the R-G-B channels correspond to bands 39-29-19, and DIF denotes the difference in band 28).

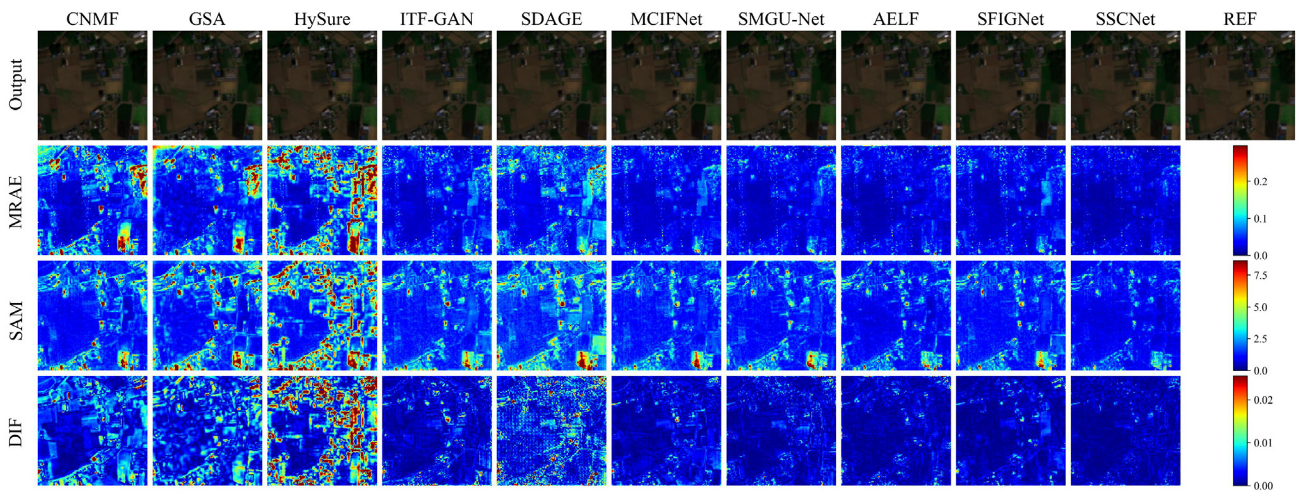

Fig. 4. Experimental results on the Chikusei dataset (the R-G-B channels correspond to bands 59-39-19, and DIF denotes the difference in band 28).

and relatively low error levels are maintained in large homogeneous regions. These results indicate that spectral distortions induced by complex urban structures can be effectively alleviated by the proposed stagewise structural-screening and progressive back-projection mechanisms.

2) Analysis of the Chikusei dataset: Chikusei scenes are primarily characterized by farmland, vegetation, and fine-grained land-cover types, with dense parcel boundaries and relatively continuous texture variations. As shown in Table III, superior performance is achieved by SSCNet in terms of PSNR, SAM, and ERGAS, indicating that farmland boundaries and the spectral continuity of vegetation regions can be effectively restored. Compared with traditional methods, edge blurring at parcel junctions is substantially reduced by SSCNet. The error maps in Fig. 4 show that weaker error diffusion at fragmented parcel boundaries and smoother boundary transitions are achieved by SSCNet. These observations demonstrate that irrelevant high-frequency texture interference can be suppressed by the proposed structural-screening mechanism, thereby improving reconstruction stability in agricultural scenes.

3) Analysis of the PaviaC dataset: PaviaC scenes are characterized by building roofs, road edges, and shadowed regions, where differences among urban materials and edge mixing are relatively pronounced. As shown in Table IV and Fig. 5, low reconstruction errors are still maintained by SSCNet under high-magnification degradation. In the DIF maps, background errors are limited, and roof contours and road textures are restored with greater continuity, demonstrating its advantages in urban edge-structure preservation and local spectral-bias suppression. Particularly under larger degradation scale factors, different urban material regions can still be effectively discriminated by SSCNet, and spectral distortions induced by edge mixing are reduced.

4) Analysis of the PaviaU dataset: PaviaU is characterized by complex urban details, such as roads, buildings, and parking-lot grid patterns, and local regions are prone to being affected by material mixing and noise perturbations. As shown

TABLE III
RESULTS ON THE CHIKUSEI DATASET

Ratio	Metrics	CNMF	GSA	HySure	ITF-GAN	SDAGE	MCIFNet	SMGU-Net	AELF	SFIGNet	SSCNet
×4	PSNR↑	41.7184	43.1256	41.8503	41.7596	39.1249	42.7925	42.9253	43.6370	43.2249	**46.1319**
	RMSE↓	1.8770	1.5962	1.8487	2.0824	2.8203	1.8489	1.8209	1.6776	1.7591	**1.2588**
	ERGAS↓	1.7926	1.4942	1.7607	1.9003	2.4440	1.7134	1.6249	1.4877	1.7271	**1.2611**
	SAM↓	1.4945	1.5525	1.5308	1.8177	2.3069	1.6847	1.6595	1.4328	1.5978	**1.1614**
	CC→1	0.9987	0.9989	0.9988	0.9985	0.9972	0.9988	0.9988	0.9990	0.9989	**0.9994**
	MoAE↓	1.1531	0.9872	1.2185	1.3497	1.9115	1.2004	1.1619	1.0151	1.1151	**0.7742**
	UIQI→1	0.9876	0.9908	0.9881	0.9848	0.9762	0.9879	0.9896	0.9914	0.9875	**0.9944**
	SSIM↑	**0.9946**	0.9937	0.9939	0.9892	0.9801	0.9910	0.9913	0.9922	0.9913	0.9941
×8	PSNR↑	37.3727	37.3541	37.4364	41.3072	38.1746	41.7778	42.0071	42.8146	41.9404	**43.7902**
	RMSE↓	3.0956	3.1022	3.0730	2.1937	3.1464	2.0780	2.0239	1.8442	2.0395	**1.6483**
	ERGAS↓	1.0917	1.0779	1.1260	0.9781	1.2922	0.9347	0.8962	0.8395	0.9361	**0.7521**
	SAM↓	1.9496	2.6063	2.1351	1.8712	2.4443	1.8198	1.8217	1.6245	1.7780	**1.4958**
	CC→1	0.9972	0.9962	0.9972	0.9984	0.9966	0.9985	0.9986	0.9988	0.9985	**0.9991**
	MoAE↓	1.8798	2.0442	1.9559	1.4018	2.1056	1.3159	1.2735	1.2294	1.2679	**1.0547**
	UIQI→1	0.9805	0.9806	0.9798	0.9832	0.9725	0.9848	0.9874	0.9891	0.9845	**0.9912**
	SSIM↑	0.9903	0.9868	0.9892	0.9890	0.9789	0.9900	0.9902	0.9901	0.9900	**0.9920**
×16	PSNR↑	26.5809	27.6405	26.5128	40.7654	37.4280	41.7782	41.6788	41.6395	41.4177	**42.6704**
	RMSE↓	10.7234	9.4919	10.8079	2.3349	3.4288	2.0779	2.1018	2.1114	2.1660	**1.8751**
	ERGAS↓	1.3298	1.1985	1.2617	0.4952	0.7723	0.4569	0.4672	0.4729	0.4773	**0.4246**
	SAM↓	4.4189	5.9432	6.4270	1.9112	2.6192	1.7565	1.7974	1.7979	1.7883	**1.6086**
	CC→1	0.9748	0.9796	0.9805	0.9982	0.9961	0.9985	0.9985	**0.9985**	0.9984	0.9981
	MoAE↓	5.9735	5.9724	6.4738	1.4861	2.3569	1.2923	1.3158	1.3356	1.3238	**1.1815**
	UIQI→1	0.9051	0.9253	0.9224	0.9833	0.9680	0.9859	**0.9967**	0.9864	0.9840	0.9886
	SSIM↑	0.9366	0.9377	0.9229	0.9889	0.9734	0.9903	0.9900	0.9895	0.9900	**0.9913**

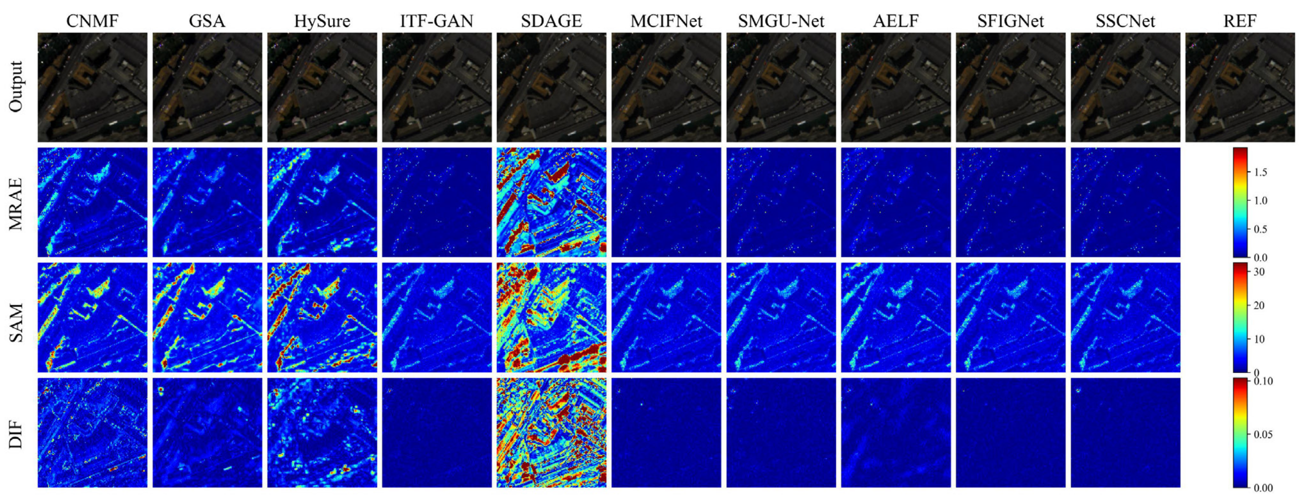

Fig. 5. Experimental results on the PaviaC dataset (the R-G-B channels correspond to bands 61-41-21, and DIF denotes the difference in band 28).

in Table V, the best performance on most metrics is achieved by SSCNet in the 4× scenario. In Fig. 6, the background regions in the error maps generated by SSCNet are almost entirely dark blue, whereas large-area high-error streaks are still observed along road edges and parking-lot grid patterns in the results of traditional methods. Under high-magnification degradation conditions, road textures and building edges can still be accurately recovered by SSCNet, while spurious textures and spectral deviations caused by excessive structural enhancement are effectively avoided.

E. Spectral Response Analysis

As shown in Fig. 7, under the 4× setting on the Chikusei dataset, the reconstructed spectra of SSCNet remain highly consistent with the reference curves across the entire spectral range. In particular, SSCNet is able to accurately capture reflectance variations in the subtle fluctuations between bands 30 and 50, as well as in the plateau region after band 80. In contrast, traditional methods generally exhibit pronounced deviations near the absorption valleys, while some deep learning models exhibit slight fluctuations in the high-band region. Furthermore, as shown in Fig. 8, the bandwise errors of SSCNet are overall closer to zero, with only small peaks observed in a few high bands, indicating that cross-band systematic bias and noise accumulation can be more effectively suppressed, thereby achieving better spectral fidelity.

F. Bandwise PSNR Comparison

As shown in Fig. 9, under the 4× setting on the Chikusei dataset, SSCNet maintains the highest overall PSNR across all bands, with its advantage being especially pronounced in the

TABLE IV
RESULTS ON THE PAVIAC DATASET

Ratio	Metrics	CNMF	GSA	HySure	ITF-GAN	SDAGE	MCIFNet	SMGU-Net	AELF	SFIGNet	SSCNet
×4	PSNR↑	39.7062	41.1641	39.6238	42.7396	40.2014	42.8124	42.3608	41.2172	42.7834	**42.9752**
	RMSE↓	2.4825	**1.5743**	2.5063	1.8602	2.4915	1.8447	1.9431	2.2166	1.8508	1.8104
	ERGAS↓	2.3293	1.9775	2.3321	1.7979	2.2691	1.7872	1.8531	2.0553	1.8021	**1.7562**
	SAM↓	3.3775	3.6604	3.5827	3.1948	3.7910	3.1931	3.2999	3.5636	3.1458	**3.0795**
	CC→1	0.9933	0.9927	0.9927	0.9949	0.9909	0.9949	0.9944	0.9927	0.9949	**0.9951**
	MoAE↓	0.9737	**0.8220**	1.5108	0.9224	1.4186	0.9234	0.9730	1.2849	0.8838	0.8566
	UIQI→1	0.9804	0.9833	0.9807	0.9854	0.9791	0.9854	0.9848	0.9823	0.9850	**0.9859**
	SSIM↑	0.9814	0.9831	0.9798	0.9830	0.9780	0.9832	0.9828	0.9808	0.9834	**0.9839**
×8	PSNR↑	34.7592	37.0270	34.5197	42.2939	39.8397	42.4274	42.2843	42.4194	42.5931	**42.8727**
	RMSE↓	4.3878	2.5347	4.5105	1.9582	2.5975	1.9283	1.9603	1.9301	1.8918	**1.8319**
	ERGAS↓	1.8755	1.4786	1.9406	0.9261	1.1820	0.9279	0.9307	0.9250	0.9063	**0.8852**
	SAM↓	4.3872	5.8167	6.3546	3.2412	3.8761	3.2969	3.3225	3.2699	3.2088	**3.1535**
	CC→1	0.9842	0.9876	0.9888	0.9944	0.9901	0.9946	0.9944	0.9946	0.9948	**0.9951**
	MoAE↓	1.7727	1.6993	3.0405	0.9850	1.4096	0.9451	0.9741	0.9738	0.9201	**0.8964**
	UIQI→1	0.9608	0.9717	0.9605	0.9854	0.9784	0.9858	0.9848	0.9854	0.9855	**0.9861**
	SSIM↑	0.9606	0.9675	0.9509	0.9827	0.9768	0.9831	0.9826	0.9828	0.9831	**0.9836**
×16	PSNR↑	25.1346	23.0676	24.4923	42.2860	39.8305	42.2857	42.0152	41.9968	42.2437	**42.5435**
	RMSE↓	13.2887	12.6443	14.3087	1.9599	2.6002	1.9600	2.0220	2.0263	1.9695	**1.9027**
	ERGAS↓	2.7594	3.2871	3.0556	0.4605	0.5835	0.4609	0.4660	0.4682	0.4579	**0.4462**
	SAM↓	8.4676	10.3024	18.5068	3.2232	3.7961	3.2226	3.3232	3.3214	3.1877	**3.1866**
	CC→1	0.8891	0.4178	0.9632	0.9947	0.9906	0.9947	0.9943	0.9943	0.9946	**0.9950**
	MoAE↓	5.3381	7.9296	9.4838	0.9746	1.3775	0.9696	1.0390	1.0263	**0.9443**	0.9616
	UIQI→1	0.7972	0.3291	0.8066	0.9863	0.9801	0.9865	0.9856	0.9860	0.9862	**0.9871**
	SSIM↑	0.8016	0.5681	0.7960	0.9829	0.9780	0.9831	0.9825	0.9826	0.9830	**0.9834**

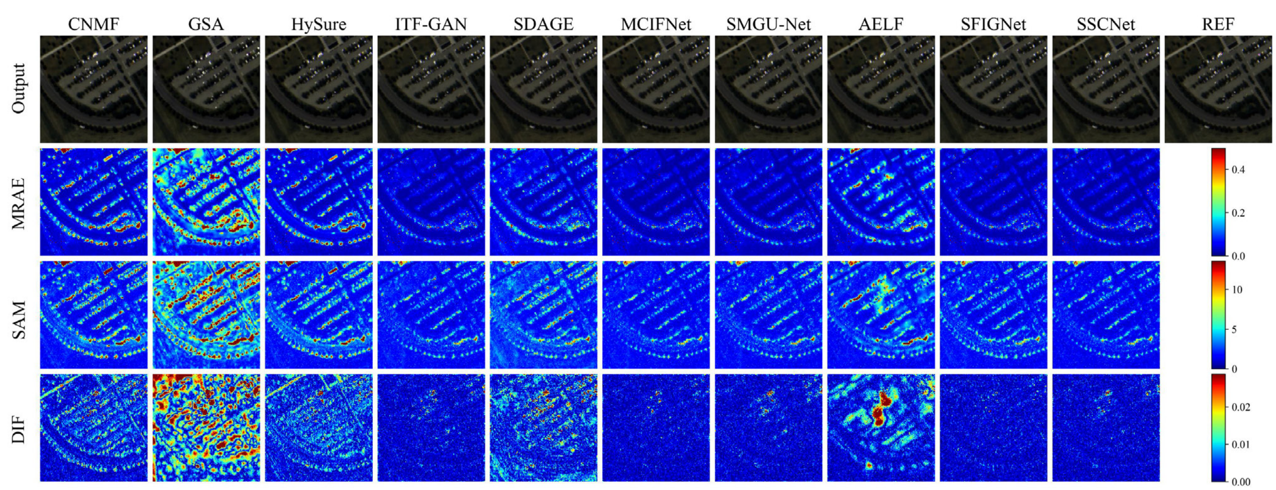

Fig. 6. Experimental results on the PaviaU dataset (the R-G-B channels correspond to bands 61-51-21, and DIF denotes the difference in band 28).

30–60 and 80–100 band ranges. Compared with traditional methods such as CNMF and GSA, SSCNet generally achieves an improvement of approximately 3–5 dB in these key spectral ranges, indicating a stronger capability for detail recovery and noise suppression in bands corresponding to complex land-cover types such as vegetation and soil. Although some deep learning models have already improved over traditional methods, they still exhibit noticeable fluctuations in the high-band region. In contrast, the SSCNet curve is smoother, and its troughs are shallower, suggesting that SSCNet contributes to improved reconstruction stability across different spectral bands.

G. Ablation Study

1) Analysis of the Performance Contributions of Individual Network Components: Table VI shows that the gains of SSCNet originate from the explicit division of labor among three components rather than from the stacking of a single module. After removing SCLM, PSNR, SSIM, and multiple spectral metrics decrease simultaneously, indicating that the HSI-dominated spectral-basis construction stage serves as the foundation for subsequent structural selection and progressive reconstruction. When FGSI or SCFI is removed, SSIM and UIQI degrade more noticeably, suggesting that structural selection and progressive back-projection play key roles in preserving spatial details and restoring structural consistency. The PSNR of the single-stage SCFI-only variant is significantly lower than that of any two-stage combination, indicating that without the spectral-basis constraint from SCLM and the structural selection from FGSI, relying only on one-shot fusion inference makes it difficult to achieve stable spectral–spatial collaborative reconstruction.

TABLE V
RESULTS ON THE PAVIAU DATASET

Ratio	Metrics	CNMF	GSA	HySure	ITF-GAN	SDAGE	MCIFNet	SMGU-Net	AELF	SFIGNet	SSCNet
	PSNR↑	37.1156	37.9761	37.4826	41.2446	38.0535	41.7720	41.3348	38.8299	41.0481	**41.8063**
	RMSE↓	2.2302	2.2198	2.2379	**2.0296**	3.1906	2.0794	2.1868	2.9177	2.2601	2.0712
	ERGAS↓	2.3113	2.1230	2.2130	1.6505	2.3406	**1.5660**	1.6796	2.0804	1.6319	1.5922
×4	SAM↓	2.8153	3.0134	2.9009	2.1319	2.8281	2.0264	2.1676	2.7356	2.2149	**1.9083**
	CC→1	0.9932	0.9934	0.9943	0.9969	0.9934	0.9972	0.9969	0.9946	0.9967	**0.9972**
	MoAE↓	1.2735	1.2278	1.2601	1.1074	1.7290	1.0260	1.0705	1.8442	1.1142	**0.9650**
	UIQI→1	0.9851	0.9854	0.9864	0.9921	0.9852	0.9928	0.9917	0.9881	0.9919	**0.9929**
	SSIM↑	0.9809	0.9810	0.9812	0.9862	0.9792	0.9874	0.9865	0.9827	0.9860	**0.9884**
	PSNR↑	32.9071	34.5073	32.4585	41.0247	38.1148	41.3764	41.2702	40.3813	40.8722	**41.6219**
	RMSE↓	3.6205	3.0112	3.8124	2.2662	3.1681	2.1763	2.2031	2.4405	2.3064	**2.1156**
	ERGAS↓	1.8048	1.4678	1.8531	0.8326	1.1448	0.8027	0.8253	0.8937	0.8180	**0.7953**
×8	SAM↓	4.3144	4.9554	5.1178	2.2218	2.8902	2.1684	2.1880	2.3789	2.3076	**2.0183**
	CC→1	0.9844	0.9884	0.9877	0.9968	0.9936	0.9970	0.9969	0.9962	0.9966	**0.9971**
	MoAE↓	2.2847	1.9128	2.5099	1.1479	1.7331	1.0894	1.1027	1.3160	1.1503	**1.0135**
	UIQI→1	0.9673	0.9747	0.9674	0.9922	0.9862	0.9926	0.9923	0.9911	0.9920	**0.9929**
	SSIM↑	0.9611	0.9668	0.9579	0.9864	0.9789	0.9875	0.9874	0.9848	0.9863	**0.9883**
	PSNR↑	22.5508	27.4800	21.5665	40.9774	37.9729	41.4654	41.2008	41.0599	40.7779	**41.5143**
	RMSE↓	11.9283	6.7626	13.3596	2.2786	3.2203	2.1541	2.2208	2.2571	2.3316	**2.1420**
	ERGAS↓	2.9809	1.7259	3.1959	0.4115	0.5849	0.3946	0.4094	0.4171	0.4118	**0.3941**
×16	SAM↓	11.3820	10.6010	17.7710	2.2799	2.9519	**2.1562**	2.3017	2.2686	2.3390	2.1574
	CC→1	0.8419	0.9603	0.9176	0.9967	0.9934	0.9971	0.9969	0.9968	0.9965	**0.9971**
	MoAE↓	7.4199	2.4964	9.6052	1.1642	1.7545	**1.0638**	1.1301	1.1404	1.1478	1.0699
	UIQI→1	0.7295	0.9105	0.7819	0.9924	0.9865	0.9930	0.9925	0.9924	0.9922	**0.9932**
	SSIM↑	0.7396	0.8550	0.6939	0.9872	0.9807	0.9882	0.9876	0.9870	0.9871	**0.9887**

TABLE VI
PERFORMANCE CONTRIBUTIONS OF DIFFERENT NETWORK COMPONENTS

SCLM	FGSI	SCFI	PSNR↑	RMSE↓	ERGAS↓	SAM↓	CC→1	MoAE↓	UIQI→1	SSIM↑
	✓	✓	45.9499	1.2854	1.3011	1.2035	0.9994	0.8150	0.9939	0.9938
✓		✓	45.7832	1.3103	1.3471	1.1889	0.9994	0.8335	0.9933	0.9935
✓	✓		45.7948	1.3086	1.3331	1.2200	0.9994	0.8290	0.9936	0.9935
		✓	44.7275	1.4797	1.4532	1.3102	0.9992	0.9458	0.9919	0.9923
✓	✓	✓	**46.1319**	**1.2588**	**1.2611**	**1.1614**	**0.9994**	**0.7742**	**0.9944**	**0.9941**

TABLE VII
EFFECTS OF INJECTING STRUCTURAL INFORMATION AT DIFFERENT STAGES

Methods	PSNR↑	RMSE↓	ERGAS↓	SAM↓	CC→1	MoAE↓	UIQI→1	SSIM↑
Early Injection	45.0374	1.4278	1.3910	1.3076	0.9993	0.9243	0.9932	0.9928
Middle Injection	45.5547	1.3453	1.3694	1.2245	0.9994	0.8403	0.9928	0.9933
Late Injection	45.7123	1.3211	1.3088	1.2152	0.9994	0.8236	0.9937	0.9936
SSCNet	**46.1319**	**1.2588**	**1.2611**	**1.1614**	**0.9994**	**0.7742**	**0.9944**	**0.9941**

2) Effects of Injecting Structural Information at Different Stages: Table VII shows that the timing of structural information injection significantly affects reconstruction performance. Early injection produces the poorest results across all metrics, suggesting that the premature introduction of structural information, before a stable spectral basis is established, may disrupt spectral–spatial reconstruction. The performance of middle injection and late injection improves progressively, with the latter achieving better results, which indicates that delayed structural involvement can mitigate the adverse effects of early coupling. Nevertheless, both strategies remain substantially inferior to SSCNet. SSCNet achieves the best performance across all metrics, suggesting that a single structural compensation stage is insufficient to fully exploit the structural priors of MSI. In contrast, the multistage progressive incorporation of selectively refined structural information is more effective in balancing spectral fidelity and spatial detail recovery.

3) Validation of the Effectiveness of the SCLM Architectural Design: In the internal ablation study of SCLM (Table VIII), the complete configuration achieves the best performance, indicating that the HSI-dominated spectral basis construction stage requires the coordinated support of multiple complementary mechanisms. When SS2D is removed, PSNR and SSIM decline most significantly, demonstrating that long-range dependency modeling is crucial for the representation of large-scale structures. The removal of SpaBlock results in the deterioration of spatial-quality-related metrics, suggesting that local spatial modeling is indispensable to detail recovery. When SpecBlock or SCR is removed, SAM and CC show more pronounced degradation, indicating that cross-band correlation modeling and channel recalibration play a key role

TABLE VIII
EFFECTIVENESS OF THE SCLM ARCHITECTURAL DESIGN

Methods	PSNR↑	RMSE↓	ERGAS↓	SAM↓	CC→1	MoAE↓	UIQI→1	SSIM↑
w/o SS2D	46.0594	1.2693	1.2804	1.1799	0.9994	0.7914	0.9941	0.9940
w/o SpaBlock	45.9250	1.2891	1.3035	1.2050	0.9994	0.8082	0.9939	0.9939
w/o SpecBlock	46.0974	1.2638	1.2743	1.1820	0.9994	0.7835	0.9942	0.9940
w/o SCR	46.0840	1.2657	1.2943	1.1765	0.9994	0.7953	0.9940	0.9940
SCLM	**46.1319**	**1.2588**	**1.2611**	**1.1614**	**0.9994**	**0.7742**	**0.9944**	**0.9941**

TABLE IX
EFFECT OF SS2D STACKING DEPTH ON LATENT-SPACE MODELING

K	PSNR↑	RMSE↓	ERGAS↓	SAM↓	CC→1	MoAE↓	UIQI→1	SSIM↑	FLOPs (G)	Model size (MB)	#Para. (M)
$K = 1$	46.1027	1.2630	1.2719	1.1776	0.9994	0.7840	0.9942	0.9941	13.28	**3.65**	0.87
$K = 2$	**46.1319**	**1.2588**	**1.2611**	**1.1614**	**0.9994**	**0.7742**	**0.9944**	**0.9941**	**13.28**	3.87	**0.87**
$K = 3$	46.1226	1.2601	1.3060	1.1760	0.9994	0.7930	0.9940	0.9939	13.28	4.09	0.87
$K = 4$	45.9845	1.2803	1.3030	1.1893	0.9994	0.8107	0.9939	0.9938	13.28	4.31	0.87

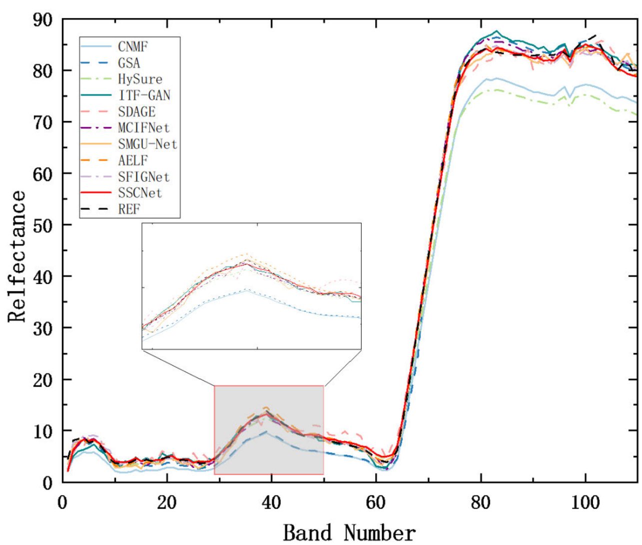

Fig. 7. Comparison of spectral curves produced by different methods.

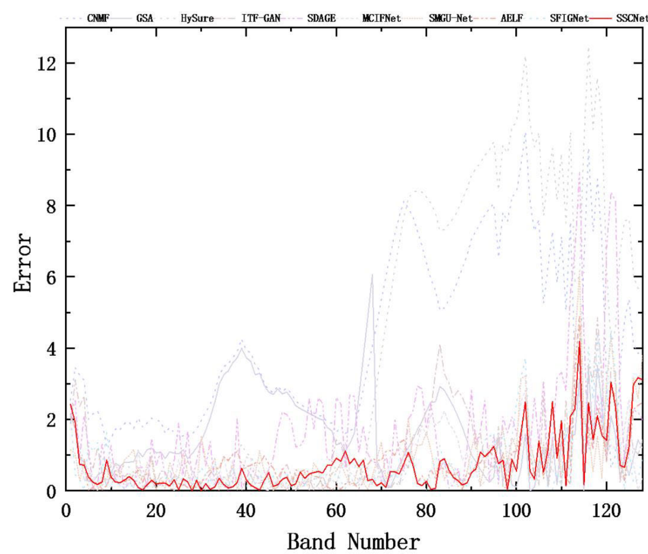

Fig. 8. Comparison of spectral errors of different methods.

in maintaining spectral consistency. Overall, SCLM provides a stable and effective latent representational foundation for subsequent reconstruction.

4) Effect of SS2D Stacking Depth on Latent-Space Modeling: Table IX further analyzes the effect of the SS2D stacking depth K in SCLM on latent-space modeling. Under the premise that the computational cost and number of parameters remain basically unchanged, all metrics first improve and then decline as K increases. When $K = 2$, metrics such as PSNR, RMSE, UIQI, and SSIM all reach their optimal values, indicating that an appropriate degree of long-range dependency modeling is beneficial for enhancing the representational capacity of the latent space and overall reconstruction quality. When $K = 1$, the long-range modeling capability is relatively weak, and the performance decreases slightly. When K increases to 3 and 4, the metrics decline instead of improving, indicating that overly deep stacking not only introduces redundant modeling but also weakens the extraction of effective contextual information. Overall, the choice of K is not merely an empirical setting, but is directly associated with the requirement of SCLM to model long-range spectral–spatial dependencies. With a moderate increase in K, a more stable HSI-dominated latent-space spectral basis can be constructed, whereas excessively deep stacking may introduce redundant contextual modeling and degrade effective feature representations.

5) Comparative Analysis of Different Structural Injection Strategies: Table X and Fig. 10 jointly validate the effectiveness of the FGSI-based structural injection strategy. As reported in Table X, FGSI achieves the best overall performance across all evaluation metrics, demonstrating its stronger ability to balance spatial detail recovery and spectral fidelity. Direct HF injection degrades the results compared with w/o structure injection, indicating that directly introducing unscreened high-frequency structures may cause additional reconstruction errors. Vanilla attention injection brings limited

TABLE X
PERFORMANCE COMPARISON OF DIFFERENT STRUCTURAL INJECTION STRATEGIES

Methods	PSNR↑	RMSE↓	ERGAS↓	SAM↓	CC→1	MoAE↓	UIQI→1	SSIM↑
w/o Structure Injection	46.0753	1.2670	1.2914	1.1753	0.9994	0.7865	0.9940	0.9940
Direct HF Injection	46.0038	1.2775	1.2958	1.1919	0.9994	0.7965	0.9939	0.9940
Vanilla Attention Injection	46.0705	1.2677	1.2765	1.1802	0.9994	0.8004	0.9942	0.9940
FGSI	**46.1319**	**1.2588**	**1.2611**	**1.1614**	**0.9994**	**0.7742**	**0.9944**	**0.9941**

TABLE XI
EFFECTIVENESS OF THE FGSI ARCHITECTURAL DESIGN

Methods	PSNR↑	RMSE↓	ERGAS↓	SAM↓	CC→1	MoAE↓	UIQI→1	SSIM↑
w/o DCT	46.0744	1.2671	1.2832	1.1858	0.9994	0.7893	0.9941	0.9940
w/o Channel Path	46.0594	1.2693	1.2837	1.1791	0.9994	0.7830	0.9940	0.9941
w/o Spatial Path	46.0898	1.2649	1.2873	1.1846	0.9994	0.7758	0.9940	0.9940
w/o Cross-modal Gate	46.0899	1.2649	1.2715	1.8281	0.9994	0.7901	0.9942	0.9941
FGSI	**46.1319**	**1.2588**	**1.2611**	**1.1614**	**0.9994**	**0.7742**	**0.9944**	**0.9941**

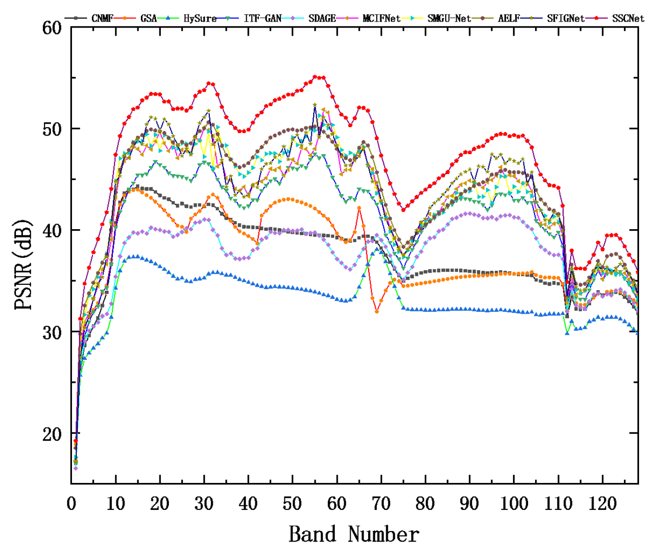

Fig. 9. Comparison of PSNR curves of different methods.

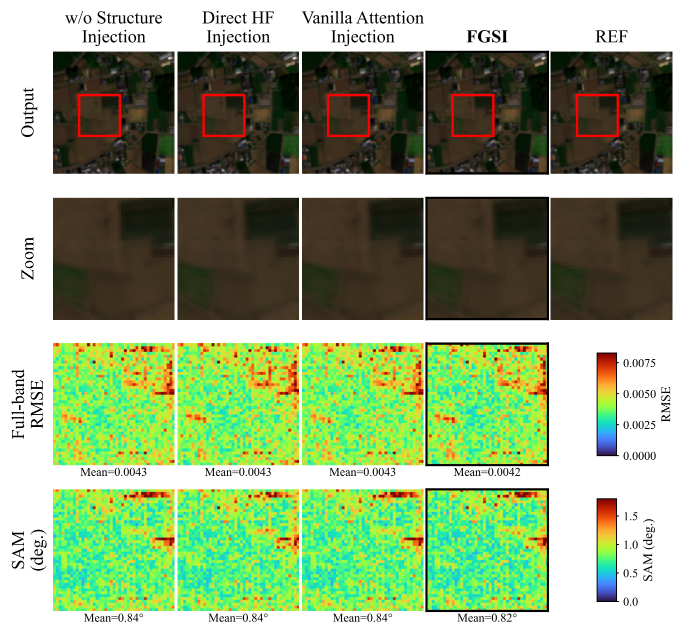

Fig. 10. Visual comparison of reconstruction results and structural errors for different structure injection strategies in a complex-texture region.

improvement, but it is still inferior to FGSI. Fig. 10 further provides local visual and error-map evidence. In the selected region, FGSI obtains the lowest full-band RMSE and SAM values, suggesting lower reconstruction error and better spectral consistency. These results confirm that frequency-guided structural screening is more reliable than direct high-frequency enhancement or conventional attention-based modulation for introducing useful MSI structural cues into HR-HSI reconstruction.

6) Analysis of the Effectiveness of the FGSI Architectural Design: In the ablation study on the architectural design of FGSI (Table XI), the complete FGSI consistently outperforms all reduced configurations across all evaluation metrics, indicating that each branch makes a positive contribution to high-frequency structural alignment and spectral preservation. When DCT is removed, ERGAS and MoAE show marked deterioration, demonstrating that high-frequency extraction in the frequency domain is crucial for suppressing noise and low-frequency interference. Removing either the channel path or the spatial path leads to slight degradation in PSNR, RMSE, and MoAE, suggesting that the two branches are complementary in modeling cross-modal high-frequency responses. When the cross-modal gate is removed, SAM increases significantly, indicating that it is particularly important for suppressing spectral distortions caused by inconsistent high-frequency injection. Overall, DCT-based frequency encoding, dual-path modeling, and cross-modal gating collectively constitute the effective fusion mechanism of FGSI.

7) Analysis of the Role of the Stagewise Coarse-to-Fine Fusion Strategy in the SCFI Module: In the ablation study of the internal submodules of SCFI (Table XII), the complete configuration achieves the best performance across all evaluation metrics, demonstrating the overall effectiveness of the coarse-to-fine fusion strategy. When AFMM is removed, all

TABLE XII
EFFECT OF THE STAGEWISE COARSE-TO-FINE FUSION STRATEGY

Methods	PSNR↑	RMSE↓	ERGAS↓	SAM↓	CC→1	MoAE↓	UIQI→1	SSIM↑
w/o CoarseEncoder	46.0899	1.2649	1.3006	1.1806	0.9994	0.7857	0.9939	0.9940
w/o AFMM	46.1167	1.2610	1.2979	1.1809	0.9994	0.7876	0.9938	0.9940
w/o SRFM	46.0074	1.2769	1.2922	1.1887	0.9994	0.7948	0.9940	0.9940
w/o Refinement	43.4200	1.7200	1.7018	1.5615	0.9990	1.0821	0.9876	0.9917
SCFI	**46.1319**	**1.2588**	**1.2611**	**1.1614**	**0.9994**	**0.7742**	**0.9944**	**0.9941**

TABLE XIII
EFFECT OF THE NUMBER OF FUSION STAGES ON MODEL PERFORMANCE

T	PSNR↑	RMSE↓	ERGAS↓	SAM↓	CC→1	MoAE↓	UIQI→1	SSIM↑	FLOPs (G)	Model size (MB)	#Para. (M)
$T = 1$	45.6479	1.3309	1.3275	1.2176	0.9994	0.8519	0.9935	0.9934	**9.4609**	**2.73**	**0.61**
$T = 2$	**46.1319**	**1.2588**	**1.2611**	**1.1614**	**0.9994**	**0.7742**	**0.9944**	**0.9941**	13.28	3.87	0.87
$T = 3$	45.9822	1.2807	1.2966	1.1954	0.9994	0.7833	0.9940	0.9940	17.10	5.01	1.12
$T = 4$	46.0596	1.2693	1.2767	1.1791	0.9994	0.7854	0.9943	0.9940	20.93	6.14	1.38

TABLE XIV
COMPARISON RESULTS WITH UNSUPERVISED METHODS

Methods	Ratio	PSNR↑	RMSE↓	ERGAS↓	SAM↓	CC→1	MoAE↓	UIQI→1	SSIM↑
EDIP-Net	×4	44.5934	1.5045	1.5648	1.2366	0.9994	0.8925	0.9931	0.9916
	×8	42.0144	2.0222	0.9964	1.6110	0.9989	1.1783	0.9890	0.9873
	×16	37.3832	3.4465	0.7747	2.3906	0.9707	2.0588	0.9738	0.9707
M2U-Net	×4	44.6023	1.5011	1.5668	1.2347	0.9994	0.8714	0.9932	0.9916
	×8	41.8778	2.0542	1.0001	1.6437	0.9989	1.1924	0.9889	0.9870
	×16	40.0511	2.5350	0.5618	1.9766	0.9983	1.4858	0.9845	0.9835
XINet	×4	43.5434	1.6958	1.9343	1.3952	0.9993	0.9840	0.9887	0.9905
	×8	41.6161	2.1171	1.0213	1.6831	0.9988	1.2251	0.9869	0.9884
	×16	39.4729	2.7095	0.5995	2.0009	0.9981	1.5374	0.9819	0.9857
SSCNet	×4	46.1319	1.2588	1.2611	1.1614	0.9994	0.7742	0.9944	0.9941
	×8	43.7902	1.6483	0.7521	1.4958	0.9991	1.0547	0.9912	0.9920
	×16	42.6704	1.8751	0.4246	1.6086	0.9981	1.1815	0.9886	0.9913

metrics show slight degradation, indicating that its adaptive regulation of structural and spectral priors contributes to improved reconstruction quality. After the removal of SRFM, PSNR and spectral error show further deterioration, suggesting that cross-stage structural refinement plays a positive role in suppressing residual distortions. When the CoarseEncoder is omitted, ERGAS and MoAE worsen significantly, indicating that the explicit encoding of coarse reconstruction and error information is beneficial for subsequent target-guided constraints. The removal of the multistage refinement leads to the most pronounced degradation in PSNR, SAM, and MoAE. Overall, these modules collectively form the reconstruction closed loop of SCFI and are critical for ensuring robust reconstruction.

8) Effect of the Number of Fusion Stages on Model Performance: Table XIII presents the effect of the number of fusion stages, denoted by T, in SCFI on performance. As the computational cost and number of parameters increase with larger T, the setting $T = 2$ achieves the best performance in terms of reconstruction error and overall quality metrics, suggesting that a two-stage fusion strategy can more effectively exploit the updated spectral and structural information. When $T = 1$, detail compensation is insufficient, and a greater amount of residual spectral error remains. When $T > 2$, performance fails to improve further and instead exhibits slight degradation. Considering both reconstruction performance and computational cost, $T = 2$ provides a more reasonable tradeoff and was therefore selected as the default number of stages in SSCNet.

H. Comparison With Unsupervised Methods

To further evaluate the reconstruction performance of SSCNet under different learning paradigms, comparative experiments are additionally conducted on the Chikusei dataset using three unsupervised HSI super-resolution methods, namely, EDIP-Net [47], M2U-Net [48], and XINet [49]. The results are reported in Table XIV. Particularly under the 16× high-magnification degradation condition, the advantage of the proposed method becomes more evident. A PSNR of 42.6704 dB is achieved by SSCNet, which is 2.6193 dB higher than that obtained by the second-best baseline method; better RMSE, ERGAS, and SAM values are also obtained. This indicates that spatial details can be effectively restored and spectral consistency can be well preserved by SSCNet, even when spatial information is severely missing. Overall, the best results on most evaluation metrics are achieved by SSCNet, demonstrating its more stable spectral–spatial reconstruction capability.

TABLE XV
ROBUSTNESS TEST RESULTS OF DIFFERENT METHODS

SNR	Metrics	CNMF	GSA	HySure	ITF-GAN	SDAGE	MCIFNet	SMGU-Net	AELF	SFIGNet	SSCNet
SNR=10	PSNR↑	23.6434	23.7508	23.6818	41.7991	38.3671	42.6548	43.1051	35.3961	43.4250	**45.9909**
	RMSE↓	20.0345	20.7070	20.5406	2.0729	3.0774	1.8784	1.7835	4.3325	1.7191	**1.2794**
	ERGAS↓	20.2627	19.9289	20.1022	1.9332	2.5106	1.7742	1.6078	2.3765	1.6838	**1.2938**
	SAM↓	20.1117	19.9172	19.7809	1.8363	2.3346	1.6882	1.6365	2.2091	1.5723	**1.1871**
	CC→1	0.8725	0.8762	0.8784	0.9985	0.9967	0.9988	0.9989	0.9934	0.9990	**0.9994**
	MoAE↓	16.7707	16.5175	16.3930	1.3587	2.1332	1.2060	1.1346	2.8366	1.0863	**0.7984**
	UIQI→1	0.6277	0.6302	0.6327	0.9840	0.9743	0.9863	0.9899	0.9771	0.9880	**0.9939**
	SSIM↑	0.2925	0.2956	0.2955	0.9889	0.9784	0.9907	0.9913	0.9740	0.9913	**0.9939**
SNR=20	PSNR↑	31.8883	31.9399	32.0523	41.9699	38.9520	42.6468	42.9723	39.9334	43.3049	**45.9655**
	RMSE↓	6.8288	6.7342	6.8034	2.0326	2.8770	1.8802	1.8110	2.5696	1.7430	**1.2831**
	ERGAS↓	6.5095	6.4780	6.5731	1.8659	2.5835	1.7871	1.6186	1.7986	1.7014	**1.2944**
	SAM↓	6.8386	6.8632	6.8279	1.8015	2.3049	1.6962	1.6544	1.6854	1.5823	**1.1893**
	CC→1	0.9841	0.9843	0.9842	0.9985	0.9971	0.9987	0.9989	0.9977	0.9980	**0.9994**
	MoAE↓	5.4231	5.3577	5.4207	1.3150	1.9952	1.2107	1.1376	1.7025	1.0969	**0.7987**
	UIQI→1	0.9082	0.9056	0.9074	0.9853	0.9747	0.9867	0.9899	0.9869	0.9876	**0.9940**
	SSIM↑	0.5804	0.5726	0.9537	0.9895	0.9802	0.9904	0.9913	0.9861	0.9913	**0.9939**
SNR=30	PSNR↑	38.8457	39.4706	38.9736	42.4001	38.6804	42.6690	43.1321	41.7427	43.0158	**46.0481**
	RMSE↓	2.9460	2.7276	2.8977	1.9343	2.9684	1.8754	1.7780	2.0864	1.8020	**1.2710**
	ERGAS↓	2.6770	2.5335	2.6823	1.7974	2.5222	1.7195	1.6202	1.6899	1.7778	**1.2729**
	SAM↓	2.6421	2.6998	2.6399	1.7337	2.3223	1.6901	1.6390	1.6988	1.6190	**1.1801**
	CC→1	0.9972	0.9974	0.9973	0.9987	0.9969	0.9988	0.9989	0.9985	0.9988	**0.9994**
	MoAE↓	2.2216	2.0770	2.2161	1.2441	2.0381	1.1834	1.1433	1.4129	1.1262	**0.7817**
	UIQI→1	0.9769	0.9779	0.9763	0.9865	0.9757	0.9877	0.9898	0.9892	0.9863	**0.9943**
	SSIM↑	0.8015	0.7932	0.7948	0.9901	0.9794	0.9908	0.9914	0.9900	0.9909	**0.9940**

I. Robustness Experiment

1) Effect of Noise on Experimental Results: To evaluate model stability against random noise perturbations, additive Gaussian noise with different intensities was added to the input observations, which can approximately characterize readout noise and random measurement errors in remote sensing sensors. The noise level was quantitatively controlled by the SNR to ensure a fair comparison among different methods. The robustness experimental results are presented in Table XV. As the noise intensity increases, the PSNR values of all methods decrease to varying degrees. Compared with traditional physical model-based methods and direct regression-based deep methods, SSCNet exhibits smaller performance degradation under low-SNR conditions, indicating more stable reconstruction capability in scenarios with noise interference and degradation mismatch. This also reflects more robust coordination between structural enhancement and spectral preservation.

2) Results on the Real Dataset: In this article, a three-band image with a spatial resolution of 0.5 m is used as the MSI, whereas an eight-band image with a spatial resolution of 2 m is used as the HSI. To construct training samples, the registered HSI and MSI are first cropped and normalized, and 1537 paired samples with a spatial size of 128 × 128 pixels are then randomly extracted for model training. In addition, a 400 × 400-pixel region is selected as the test area. The QNR [50] metric is adopted to comprehensively assess the spatial and spectral quality of the fusion results, with values closer to 1 indicating better fusion quality.

Fig. 11 presents the fusion results obtained by SSCNet on real cross-sensor data. It can be observed that road contours, building boundaries, and local textures from the HR-MSI are well preserved in the fused image, while an overall color distribution consistent with the LR-HSI is maintained. No obvious abnormal color shift, over-sharpening, or spurious texture can be observed. From the perspective of no-reference evaluation, a QNR value of 0.9914 is achieved by the final fusion result, indicating that favorable overall fusion quality can be obtained by SSCNet on real data.

Fig. 11. Results of SSCNet on the real dataset.

J. Complexity Comparison

Table XVI reports the model complexity and inference efficiency of different deep learning-based methods under identical testing conditions. SSCNet has only 0.87-M parameters, a model size of 3.87 MB, and 13.28-G FLOPs, indicating that its overall model scale and theoretical computational cost remain relatively low. Although lower values are obtained by MCIFNet and SFIGNet for some complexity metrics, their inference times are 138.49 and 196.85 ms, respectively, with frames per second (FPS) values of only 7 and 5, indicating relatively limited practical inference efficiency. In contrast, an inference time of 13.38 ms and an FPS of 74 are achieved by SSCNet, demonstrating favorable practical inference efficiency with low computational overhead and reflecting a better performance-complexity tradeoff.

TABLE XVI
COMPLEXITY COMPARISON OF DIFFERENT DEEP LEARNING METHODS

Metric	ITF-GAN	SDAGE	MCIFNet	SMGU-Net	AELF	SFIGNet	SSCNet
#Para. (M)	1.35	11.88	0.61	9.49	6.98	0.42	0.87
Model size (MB)	5.15	48.36	3.45	36.23	26.74	1.64	3.87
FLOPs (G)	92.93	37.03	10.09	293.69	34.00	16.34	13.28
Infer. Time (ms)	77.74	20.45	138.49	49.89	7.27	196.85	13.38
FPS	12	48	7	20	137	5	74

V. CONCLUSION

This study addresses the limitations of existing HMIF methods, in which spectral modeling and structural compensation are often synchronously coupled, and the utilization of structural information lacks stagewise organization, by proposing an SSCNet. Unlike fusion strategies that directly strengthen synchronous cross-modal interaction, HR-HSI reconstruction is decomposed by SSCNet into three successive stages: spectral basis construction, cross-modal structural screening, and progressive fusion inference. Experimental results on multiple datasets demonstrate that SSCNet achieves competitive reconstruction performance under different degradation scales, while ablation and robustness studies further validate the effectiveness of the proposed stagewise collaborative design. It should be noted that the experiments are mainly conducted under simulated degradation conditions following Wald's protocol. In future work, complex factors such as registration errors and degradation-model mismatch in real cross-sensor scenarios will be further incorporated to enhance the robustness and generalization capability of SSCNet in practical remote sensing applications.

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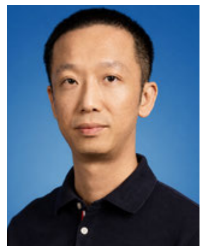
Ruiping Liu received the master's degree from the School of Electronics, Information and Optical Engineering, Taiyuan University of Technology, Taiyuan, Shanxi, China, in 2024. She is currently pursuing the Ph.D. degree with the School of Information and Communication Engineering, Xi'an Jiaotong University, Xi'an, China.

Her research interests are primarily focused on hyperspectral image fusion and object detection.



Jie Li received the B.S. degree in physics and the M.S. and Ph.D. degrees in optical sciences and electronic science and technology from Xi'an Jiaotong University, Xi'an, China, in 2006, 2009, and 2012, respectively.

He is currently a Professor with the School of Information and Communications Engineering, Xi'an Jiaotong University. His research interests include intelligent image processing and interpretation, computational optics, visible and infrared imaging spectrometry, polarimetry, and interferometry. Since graduation, he has been primarily engaged in research on spectral imaging and polarization imaging.

Pan Wang received the bachelor's and master's degrees from the School of Automation and Information Engineering, Xi'an University of Technology, Xi'an, Shaanxi, China, in 2012 and 2015, respectively.

He is currently a Post-Doctoral Researcher and an Assistant Professor with the School of Information and Communication Engineering, Xi'an Jiaotong University, Xi'an. His research interests are primarily focused on compressive spectral imaging and computational imaging.

Chunyu Zhu received the master's and Ph.D. degrees from the College of Geo-Exploration Science and Technology, Jilin University, Changchun, China, in 2020 and 2024, respectively.

He is currently a Post-Doctoral Researcher at Hangzhou Institute of Technology, Xidian University, Hangzhou, China. His research interests include multisource image fusion, object detection, and urban change detection methods.